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Final Year Project Report

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BACHELOR'S DEGREE IN COMPUTER SCIENCE

Title:

**Design and Development of a Workflow for Managing
Application Access Rights in the Information
System of the Bank**

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I would like to dedicate this work to all those who are dear to me.

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General Introduction

Despite the rapid digitization that has been underway in various industries, the banking industry continues to have a significant reliance on rigid, manual processes for employee access governance and internal audit functions. Access management in the majority of banks continues to be based on disconnected paper-based systems, approval workflow in silos, and unstructured communication, and therefore results in a laborious but very error-prone and blind-spot-filled process.

This antiquated model is critically flawed: approvals can be delayed as requests are misdirected or not signed, unauthorized modifications go unmonitored, and there is no audit trail centrally available to dispute traceability or accountability. These inefficiencies have a day-to-day impact as well as subjecting organizations to serious risk of compliance breaches, data loss, and insider threat risks with special vigor in the banking sector, where regulatory demands and security expectations are among the most demanding.

What is largely lacking in most financial institutions' existing deployments, however, is the integration of smart automation features such as Model Context Protocols (MCPs) [1] or AI-augmented auditing platforms. Whereas passive, static architectures only log activity, MCPs offer an active, real-time, context-enhanced layer that can examine and organize system activity into useful, query-able data structures. Such capabilities, however, are sadly under deployed within financial institutions, leaving compliance groups to dig through massive volumes of raw logs without sense-making tools.

Similarly, AI is substantially underleveraged in audit and access governance. While banks may use AI for fraud detection or customer analytics, its internal operational control capabilities i.e., detecting unusual access patterns, answering audit questions, or predicting escalation bottlenecks remain underleveraged. This technology gap prevents the industry from revamping its internal defense mechanisms and keeping up with an expanding digital risk environment.

There is a pressing need to shift toward AI-driven, context-aware systems to ensure secure, traceable access management. As regulatory demands grow, banks must rethink and modernize their internal security and audit processes to stay protected and compliant.

Chapter 1: Project context

1. Introduction

In this chapter, we will explore the structure and methodology of our work. We begin by presenting the hosting company, **Banque de l'Habitat (BH Bank)**, and its role in the financial sector. Then, we will analyze the current state of access management within the bank, identifying the key challenges and inefficiencies of the existing system. Based on this analysis, we will propose a digital solution designed to streamline and secure the authorization process. Finally, we will detail the **development methodology** adopted in this project.

2. Hosting organism

The work presented in this report focuses on a solution for authorization management within the internal information system applications of “*Banque de l'Habitat*” (BH).



Figure 1: Logo BH.

BH Bank is a leading player in Tunisia's financial sector since 1989, with an inherent mission to serve clients primarily housing finance, coupled with a wide range of tailored banking solutions. Committed to innovation, the bank willingly embraces digital transformation to offer improved customer experience through its digital and mobile platforms. It has Retail, Corporate, Mortgage, and Investment Banking and Digital Banking solutions, ATM and cash services, and international banking support.

3. Preliminary study

This section presents a focused comparison of leading IAM solutions based on criteria drawn from real-world banking needs and observed gaps. The evaluation highlights how current tools align with modern expectations in usability, compliance, and system adaptability. These insights form the foundation for our proposed solution.

a) Evaluation Criteria

Each criterion below represents a critical feature or requirement derived from both our study and practical gaps observed in existing IAM tools.

Chapter 1: Project context

Table 1: Chosen Criteria.

Criterion	Definition
AI & MCP Integration	Enables natural language queries, contextual audit summarization, and intelligent analysis of log files (e.g., CSV), powered by the Model Context Protocol to enhance traceability and compliance.
Ease of Use	The system is intuitive and requires no prior training, lowering the adoption barrier for employees and managers.
Banking-Specific Fit	Tailored features and workflows that align with banking terminology, structure, and regulatory compliance needs .
Audit Support	Built-in logging, traceability, approval chains, and CSV summarization to support security audits and compliance effortlessly.
Command Customization & Legacy Support	Ability to generate execution-ready files (SQL, Bash) for manual or hybrid environments, ensuring compatibility with legacy banking systems .
Modern Architecture	Built with modern frameworks (Next.js, Prisma, Tailwind, etc.), ensuring long-term maintainability, responsiveness, and seamless integration with other digital tools.

b) Analysis of existing systems

➤ Comparative study

This step will be a direct comparison of the Top current solutions on the market and the solution we will be discussing based on the criteria.

Table 2: Direct Comparison of Current Solutions.

Criterion	AI & MCP Integration	Ease of Use	Banking-Specific Fit	Audit Support	Command Customization & Legacy Support	Modern Architecture
MIM (Microsoft's on-prem IAM tool with strong AD integration) [2]	Non	Non	Non	⚠	⚠	Non
SailPoint (Enterprise-grade IAM platform focused on governance and compliance) [3]	Non	⚠	⚠	✓	Non	⚠
Okta (Cloud-based IAM service known for SSO and third-party app integration) [4]	Non	✓	Non	✓	Non	✓

➤ Synthesis

The comparison reveals critical shortcomings in existing IAM tools, particularly in AI and MCP integration, banking fit, and legacy support. While some offer partial strengths, none deliver a complete solution for the financial sector. This underscores the need for a purpose-built, modern IAM platform tailored to banking environments.

4. Problematic

The problem that we have identified in our review of the existing systems is as follows: How can we develop a technological platform that goes beyond current access management solutions by simplifying complexity, enhancing traceability, and addressing the specific needs of banks while integrating a Model Context Protocol (MCP) to support intelligent, secure auditability and operational efficiency?

Overall, these complex and interconnected challenges highlight the urgent need to rethink and reinvent access management platforms making them more streamlined, secure, and tailored to the banking sector. The goal is to ensure easier, faster, and more accountable access governance, regardless of an institution's digital maturity or internal resources.

5. Proposed solution

Our proposed solution is an AI-powered, centralized role and access management platform uniquely tailored to the bank's operational and security needs. The integrated assistant, **Auditron**, acts as a Model-Context Operator (MCO) powered by a custom Model Context Protocol (MCP), available to **Admins** and **RSSI (Information system security manager)**, supports real-time monitoring of activity, assists the audit processes, and helps interpret system logs greatly accelerating and simplifying auditors' workloads. The platform manages authorizations, permissions, audit trails, and KPI monitoring, with a secure, scalable option to the fault-prone, paper-based system in use.

6. Goals

- ✓ **Streamline access requests:** Digitalize and standardize the process of requesting and approving access across all applications.

Chapter 1: Project context

- ✓ **Enhance security and auditability:** Ensure that every action is logged, creating a clear audit trail to support compliance and internal reviews.
- ✓ **MCP-powered insights:** Use a context-driven protocol to monitor activities, runs specific queries to retrieve wanted data, and summarizes uploaded CSV files to help understand exported logs and audits.
- ✓ **Efficient authorization management:** Streamline approval chains, reduce delays, and ensure timely notifications by enforcing structured, role-based processes, with email alerts for key actions and approvals.
- ✓ **Centralize user and permission management:** Maintain unified control over user profiles and access rights across all systems.
- ✓ **Scalable and secure architecture:** Provide a modern, secure alternative to fragmented or paper-based systems, with the flexibility to grow alongside the bank's needs.

7. Actors identification

The actors whom will be interacting with my system are:

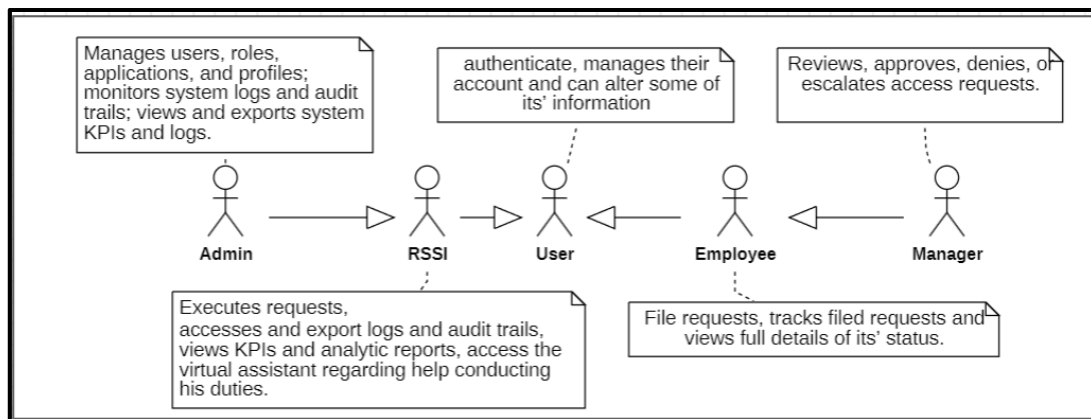


Figure 2: Actors hierarchy architecture.

8. Requirements specification

This project focuses on the analysis, design, and implementation of a centralized role and access management system tailored for the banking sector. The primary objective is to replace the existing manual and paper-based authorization process with a secure, digital platform that automates user access requests, approvals, and audit logging. The system will offer a user-

Chapter 1: Project context

friendly interface for employees and managers, support multi-level approval chains, and ensure full traceability of actions for compliance and security audits. By integrating with existing directory services and legacy systems, the platform aims to enhance operational efficiency while meeting regulatory standards.

a) Functional Requirements

Functional requirements define what the system must do to fulfill its purpose.

- User Authentication & Authorization: The system must support role-based access control (RBAC).
- User Role & Profile Management: The system should enable role assignment and edit, manage profiles related to the application the system manages, ensure access privileges are defined by these roles.
- Access Request & Approval Workflow: The system should allow users to submit access requests, notify managers for approval, and update users on request status. Approved requests are authorized after RSSI review.
- KPIs, Audit & Logging: The system must log all key activities and provide on-demand access to reports, audits, and KPIs for monitoring and analysis.
- Application Management: The system should support managing applications and integrating multiple apps for centralized authorization control.
- Security & Compliance: The system includes **Auditron**, a Model-Context Operator (MCO) for Admins and RSSIs that enables data queries, platform insights, decision support, and interactive guidance.

b) Non-functional Requirements

- Scalability: Supports large volumes of users, applications, and profiles without impacting performance.
- Performance: Efficiently processes multiple concurrent access requests with smooth user interactions.
- Security: Encrypts sensitive data in transit and at rest; enforces strict role-based access control.
- Availability: Designed to be available 24h/24h, 7d/7d and accessible while needed by all users.
- Maintainability: Modular architecture and clear documentation enable easy updates and fast administrator onboarding.

Chapter 1: Project context

- Usability: Intuitive interface with contextual tooltips across the platform with no prior training needed.
- Auditability & Compliance: Automatically logs key actions and retains records to meet security and compliance needs

9. Justification of Methodological and Technical Choices

This section justifies the selection of Agile (Scrum) and UML.

a) Project management methodology

Project management methodologies play a crucial role in guiding each phase of the project, ensuring structured planning, efficient execution, and successful completion. These methodologies provide a framework for managing resources, timelines, and risks, enabling seamless coordination throughout the project's lifecycle.

➤ Methodology choice

Waterfall is a linear approach for fixed requirements, while Agile is flexible and iterative, ideal for projects with evolving needs. Agile enables faster delivery, adaptability, and continuous improvement, making it perfect for dynamic, user-centered solutions.

➤ Agile methodologies

Although there are different design methodologies, agile methods are the most suitable because they allow for easy modifications at each stage of the process. Indeed, agile methods are iterative and incremental development approaches.

Table 3: Agile methodologies.

Methodology	Best for	Advantages	Disadvantages
Scrum	Complex projects with evolving requirements (e.g., software development).	High collaboration, structured framework, fast feedback loops.	Can be difficult to scale, requires a dedicated Scrum Master, strict rules.
Kanban	Projects with ongoing work and a need for continuous delivery (e.g., IT support, maintenance)	Simple, flexible, highly visual, encourages continuous improvement.	Lack of structure can make it hard to manage complex projects, no defined iterations.
Extreme Programming (XP)	Projects with high technical complexity and the need for frequent releases.	Emphasizes quality, early bug detection, promotes teamwork.	Requires strong technical skills, may not be suitable for large teams, requires high discipline.

➤ Opted Agile methodology

I chose Scrum for its iterative, sprint-based approach that supports continuous improvement and adaptability. With Mr. Hatem as Product Owner, priorities stayed aligned,

Chapter 1: Project context

and as the sole developer, I benefited from Scrum's structure to manage tasks and deliver steady progress.

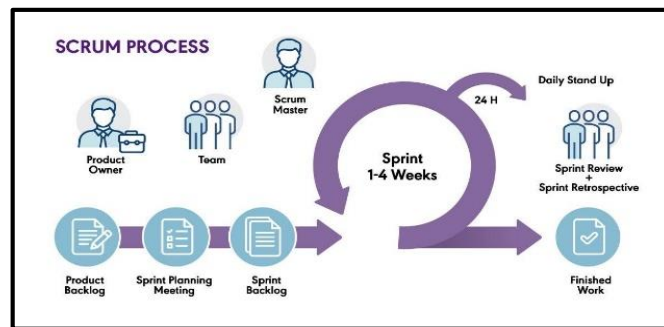


Figure 3: SCRUM methodology

➤ SCRUM organization

Table 4: Project Actors.

Role	Mission	Actors
Product Owner	Identify functionalities to develop	Mr;Hatem Mhenni
Scrum Master	Ensures the development progress	Mr;Hatem Mhenni
Team	Creates and delivers the project adopting Agile practices	Amir SMATI

b) Modeling language

➤ Modeling language Choice

Modeling languages like UML and BPMN visually represent systems and processes, UML for software structure and behavior, and BPMN for clear, consistent mapping of business workflows.

➤ Opted modeling language

I chose UML for its clear, standardized visuals that aid communication and understanding among stakeholders. Though not required by Scrum, it enhances transparency in the development process.

10. Conclusion

During this chapter, we were able to understand the general context of the project by analyzing the existing situation and then selecting the project management methodology that will organize the various phases of this project.

Chapter 2

Sprint 0: Project planning phase

1. Introduction

By analyzing the current state, I identified the functional requirements, non-functional requirements, and the stakeholders involved in the system. To optimize the efficiency of our project, I have divided its implementation into seven different sprints.

2. Product Backlog

Following the SCRUM method, the project is divided into features called User Stories. The Product Backlog is therefore composed of a list of these features to be implemented in order to finalize the product.

The table below provides an overview of the Product Backlog, presenting the functional requirements in the form of User Stories.

Table 5: Product Backlog

	Epic	ID	User Stories	Priority
1	Authentication	1.1	As a User, I should be able to log in using my credentials so that I can access my role-specific dashboard.	100
		1.2	As a User, I should be able to reset my password if I have lost it so that I can access the application. As a Admin, I should be able to add, modify, and soft delete Users so that I can manage system access securely	97
		1.3	As a Admin, I should be able to add, modify, and soft delete Users so that I can manage system access securely.	96
2	Management	2.1	As a Admin, I want to assign roles to Users so that they have appropriate permissions.	90
		2.2	As a Admin I want to impersonate any user (view the app as them) So that I can troubleshoot issues, verify permissions, assist with updates or assist with support more effectively.	87
		2.3	As a Admin, I want to add, configure each application (e.g., set commands,	86

Chapter 2. Sprint 0: Project Planning Phase

			update its information) and soft-delete so that the system stays up-to-date.	
		2.4	As a Admin, I want to define and manage profiles (e.g., role templates) so that permissions can be standardized.	85
		2.5	As an Employee, I want to view my assigned profiles, applications, and request count so that I can track my workload and my given permissions.	84
		2.6	As a Manager, I want to view my employees requests count and information so that I can oversee access approvals.	83
3	Request handling	3.1	As a Manager, I want to review, approve, reject, or escalate access requests so that access is granted responsibly.	77
		3.2	As an Employee/Manager, I want to request access to applications so that I can perform my duties.	76
		3.3	As an Employee/Manager, I want to track my request status so that I know when access is granted.	75
		3.4	As a Manager, I want to receive email notifications for new requests so that I can act promptly.	74
		3.5	As an Employee/Manager, I want to receive email notifications when my request is approved/rejected so that I'm informed.	74
		3.6	As an RSSI/Admin, I want to view all requests and their history so that I can audit access changes.	73
		3.7	As an RSSI/Admin, I want to execute approved requests (e.g., provisioning access) so that workflows are completed.	72
4	Control	4.1	As an RSSI/Admin, I want to access logs and audit trails so that I can monitor system activity.	68
		4.2	As an RSSI/Admin, I want to view KPIs & generate reports so that I can analyze trends.	68
		4.3	As an RSSI/Admin, I want to export logs/audits (CSV/PDF) so that I can share compliance evidence.	64
		4.4	As an RSSI/Admin, I want to query the virtual assistant for app-related	64

Chapter 2. Sprint 0: Project Planning Phase

			information so that I can get quick answers.	
		4.5	As an RSSI/Admin, I want to summarize CSV audit logs via the virtual assistant so that I can analyze data faster.	63

3. Sprint planning

The project plan outlines sprints with specific tasks for developing and refining the role and access management platform, tracking progress and enabling adjustments.

Table 6: Sprint plan.

Sprint 1 : Authentication & User creation	Sprint 2 : Management	Sprint 3 : Request handling	Sprint 4 : Control
• Login • Information update • Password reset • User creation	• User Management • Application Management • Profile Management	• Request Management • Request Execution	• Logs & Audit trails consulting • KPIs consulting • AudtiTron (MCO) implementation

4. Sprints duration

In this section we will describe the duration each sprint will take using Gantt diagram.

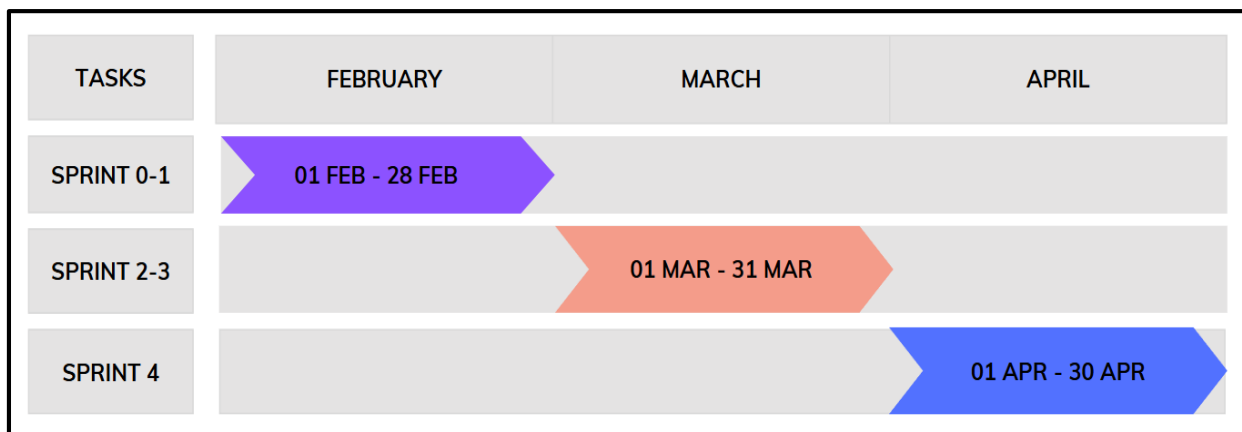


Figure 4: Gantt Diagram.

5. General use case diagram

Below, you will find the global use case diagram of our application:

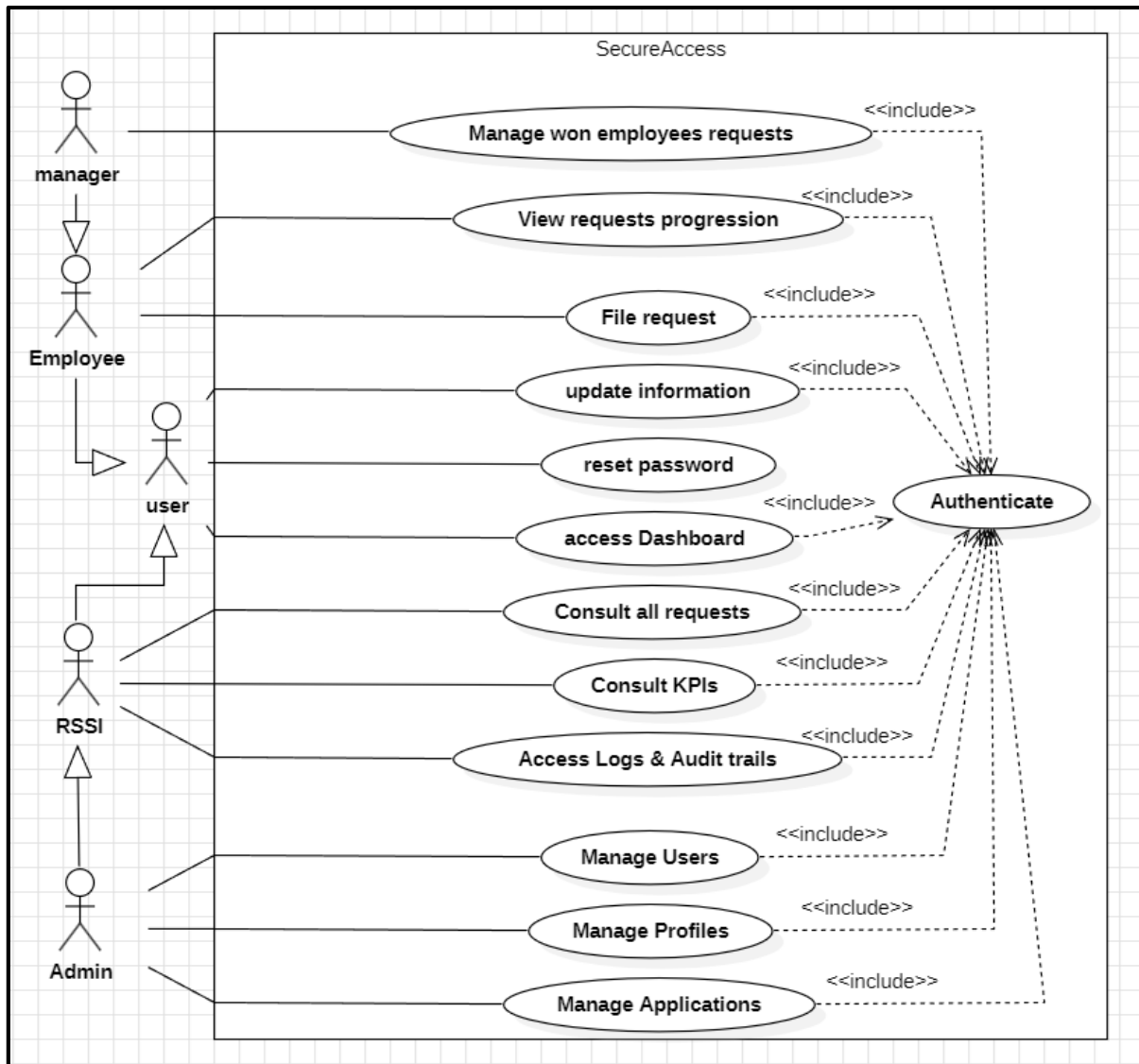


Figure 5: General use case diagram

6. Development environment

In this section we will discuss the used Software to successfully develop this system

Table 7: Software Environment.

Logo	Description
	Visual Studio Code is a lightweight code editor developed by Microsoft that runs on your desktop. It is available for Windows, macOS, and Linux. [5]

	Node.js is an open-source platform designed to run server-side applications. It operates on a single thread and is compatible with multiple platforms. [6]
	PostgreSQL is an open-source relational database management system that uses SQL to manage and store data. It is known for its reliability, scalability, and support for complex queries. [7]
	Next.js is a React-based framework for building server-side rendered and static web applications. It simplifies development with features like automatic code splitting, routing, and server-side rendering. [8]
	ThunderClient is a lightweight REST API client for testing and interacting with APIs, integrated directly into Visual Studio Code for a seamless development experience. [9]
	DBeaver is a free, multi-platform database management tool that supports various databases, including PostgreSQL. It provides a graphical interface for running queries, managing data, and browsing database structures, making it easier for developers and database administrators to work with databases efficiently. [10]
	ShadcnUI is a customizable React component library built using Tailwind CSS. It offers accessible, unstyled components designed for full flexibility. Developers can style components to match their design systems easily. It's ideal for building modern, fast, and clean UIs with full control. [11]
	TypeScript is an open-source programming language developed by Microsoft. It adds static typing to JavaScript, enhancing code quality and scalability, it also compiles to plain JavaScript, making it compatible with all browsers. It's widely used for building large, maintainable front-end and back-end applications. [12]

7. Conclusion

The functional and non-functional requirements of my project have been defined in this chapter, along with the roles of the various stakeholders involved. I have also developed a use case diagram to provide a clearer vision of the upcoming tasks.

Chapter 3

Sprint 1: Authentication & User creation (Login, Information update, Password reset, User creation)

1. Introduction

For this chapter, we carefully planned the sprint by defining the user stories that we will develop throughout this module. This organization will allow us to effectively structure our work and ensure a coherent and methodical progression in the development of our project.

2. Sprint Backlog

As the sole developer on the project, I begin each sprint with a clear planning session where I identify and prioritize the most essential features from the product backlog. These selected tasks are then developed, tested, and finalized by the end of the sprint.

The table under presents the **sprint backlog** for this cycle.

- **Sprint Start Date:** February 10, 2025.
- **Sprint End Date:** February 14, 2025.
- **Work Scale:** One working day (out of Four) equals 9 hours of work (from 10:00 AM to 8:00 PM).
- **Estimated Time:** 45 hours.
- **Sprint Objective:** Implement User creation for Admins, and authentication as well as password resetting via email for all Users.

Table 8: Sprint Backlog "Authentication & User creation ".

ID U.S	User Story	Tasks
1	As a User, I want and log in using my credentials so that I can access my dashboard and change my information.	1.1 Prepare the necessary diagrams. 1.2 Develop the use case. 1.3 Test the use case.

Chapter 3. Sprint 1: Authentication & User creation

2	As a User, I should be able to reset my password if I have lost it.	1.1 Prepare the necessary diagrams. 1.2 Develop the use case. 1.3 Test the use case.
3	As a Admin, I should be able to add, modify, and soft delete Users so that I can manage system access securely.	1.1 Prepare the necessary diagrams. 1.2 Develop the use case. 1.3 Test the use case.

3. Requirements specification

In this part we exhibit the use case diagram for this sprint.

a) Use case diagram

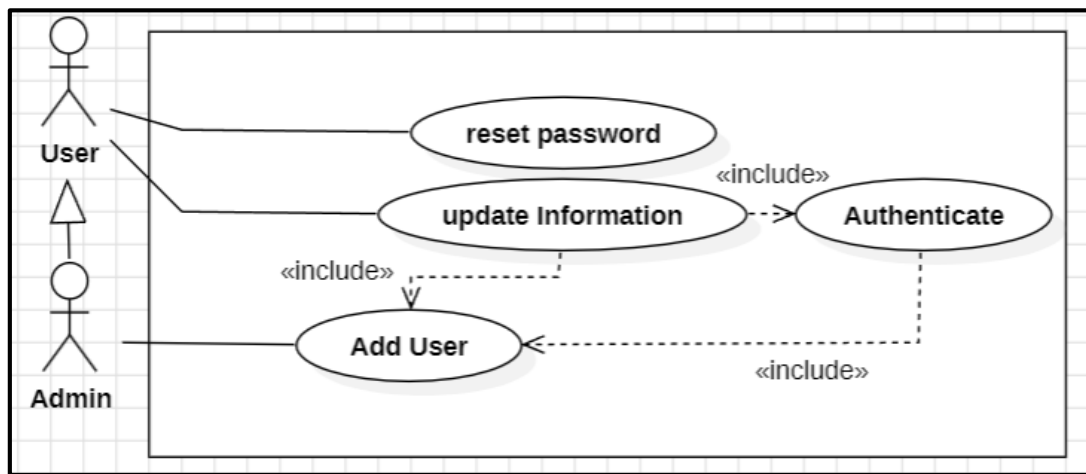


Figure 6: Sprint 1 Use Case Diagram.

b) Textual description

The following table, displays a detailed textual description for the sprint “Authentication & User creation”.

Table 9: Textual Description of UC "Authentication & User creation".

Use case	Authentication & User creation
Actors	User, Employee, Manager, Admin, RSSI
Objective	Allow Admins to create Users, and each user type to log in to the application using their credentials and access a role-specific dashboard and change their information.
Pre-condition	User navigates to home page and then login page.
Post condition	• User gets created and assigned a role, login, and a password. • If the login is successful, the user gets directed to their role-specific dashboard. • If the user forgot their password, a reset link is sent and allows them to define a new one.
Nominal Scenario	1. Admin navigates to “Add user” page, and fills in the form to add a new a user.

	2. User navigates to the “Home” page and then accesses the “Login” page. 3. System displays the login form. 4. User enters their credentials. 5. System validates their credentials. 6. User gets directed into their role-specific dashboard.
Alternative Scenario	1. Empty Fields/Inactive Account/ Inexistent Account/ Incorrect credentials: • The system returns: Error Toast • Return to step 3. • 2. Forgotten Password: • User clicks "Forgot Password?" on the login page. • System displays the reset password form. • User enters their registered email address. • System checks if the email exists: i) If not: display "User not found. Check your email and try again." ii) If found: (1) System generates a reset token. (2) Sends an email with a password reset link. (3) Displays a confirmation message (4) User follows the link and accesses the password reset form. (5) User defines a new password. (6) System updates the password and confirms reset success. (7) User can now return to login and access the app.

4. Planning

In this section, we’ll be exhibiting the **Sequence** diagram as well as the **Class** diagram that was used to achieve this Sprint.

a) Sequence diagram

The following figure, presents the Sequence diagram for this sprint.

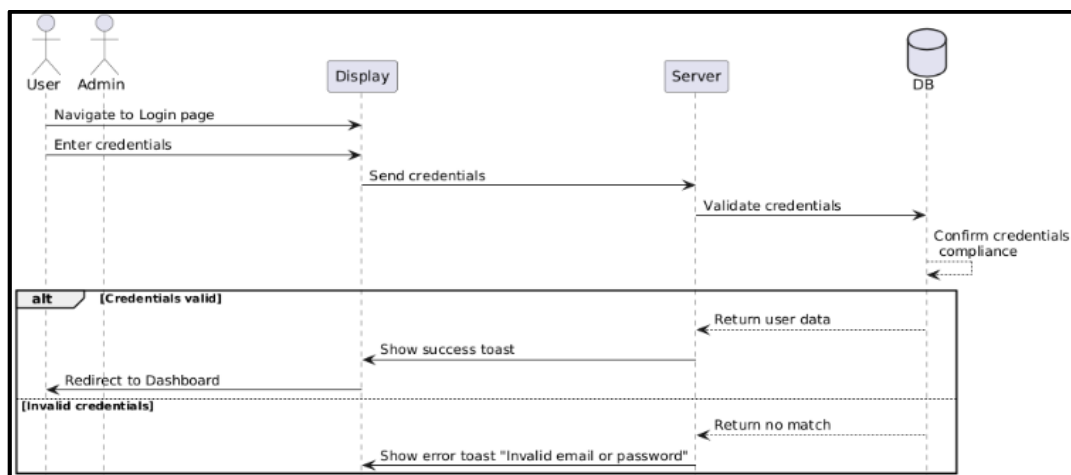


Figure 7: Sequence Diagram of Use Case "Authentication".

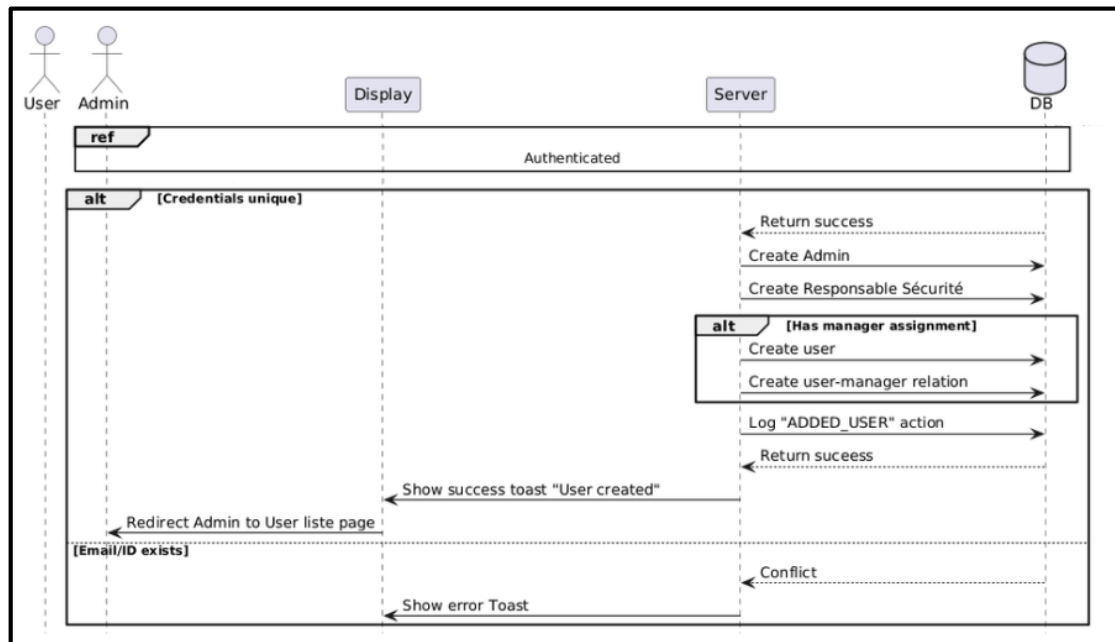


Figure 8: Sequence Diagram of Use Case "Add User".

b) Class diagram

The following figure, presents the Class diagram for this sprint.

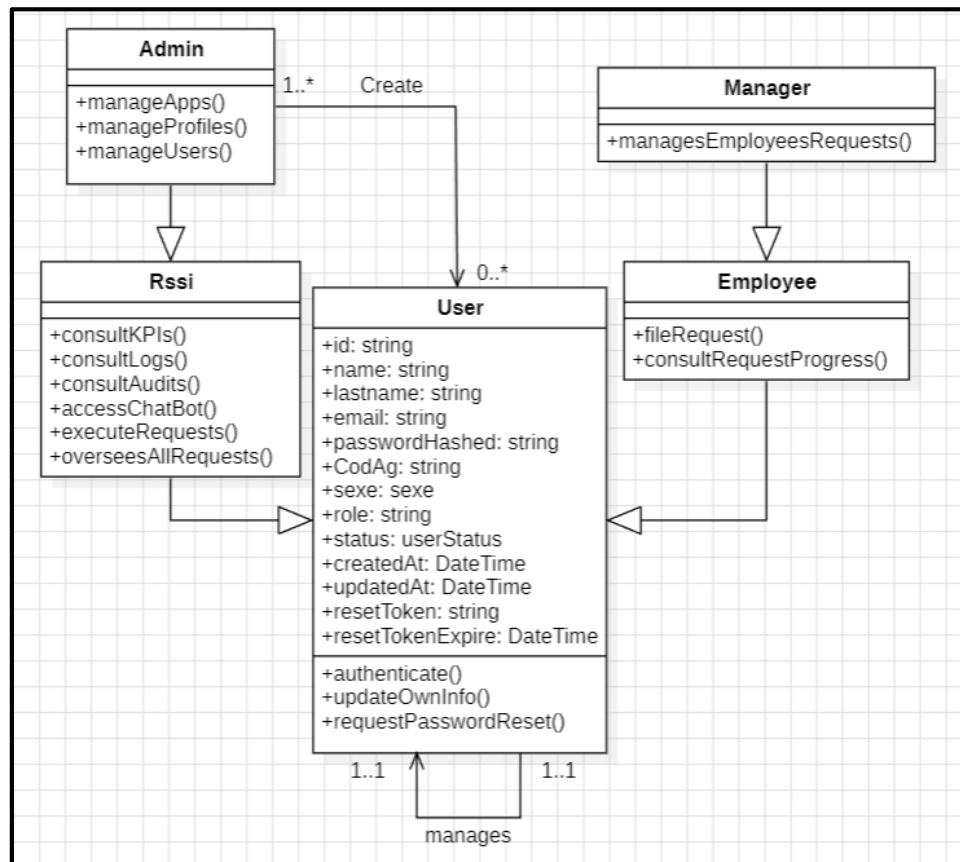


Figure 9: Class Diagram of Use Case "Authentication".

5. Realization

The following figures show some of the Pages that are present in this sprint, the rest are in the Annex section.

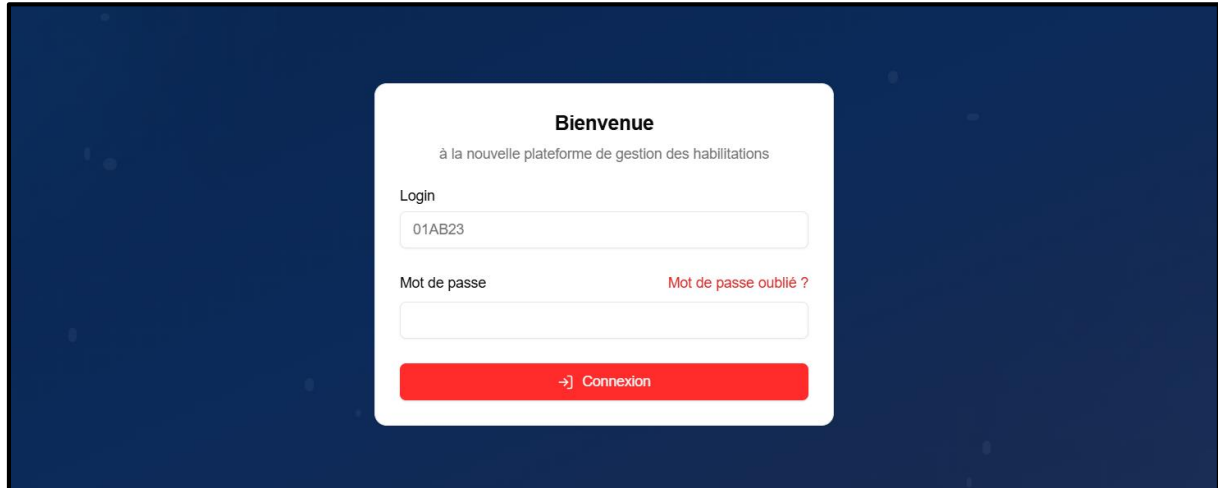


Figure 10: Login Page.

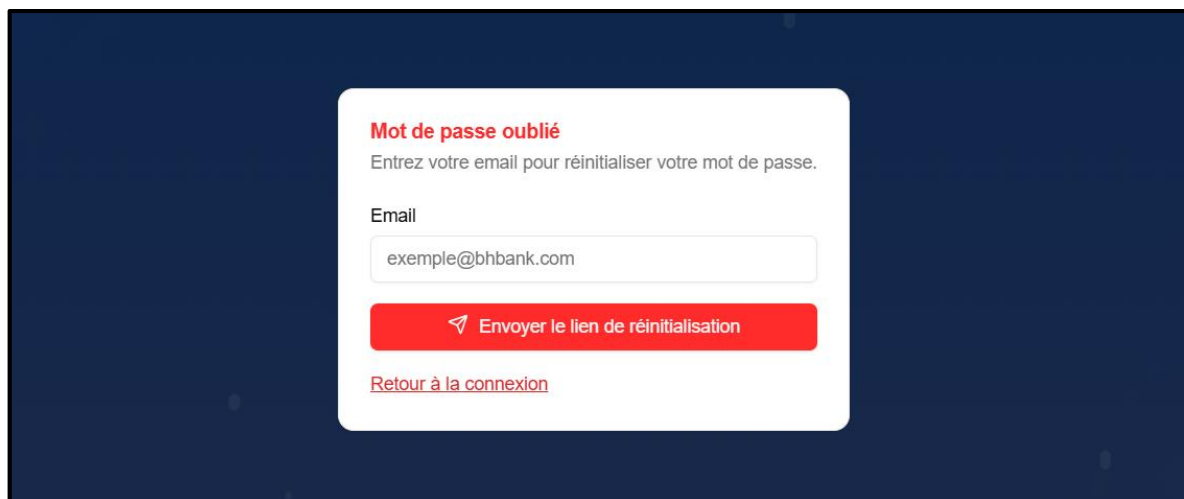


Figure 11: Forgot Password Page.

6. Conclusion

After conducting several meetings and revisions, our supervisor validated this sprint's functionalities, which focused on login, password reset, and user addition, allowing us to proceed confidently to the next phase.

Chapter 4

Sprint 2: Management (User management, Application management, Profile management).

1. Introduction

In this chapter, we planned the sprint by defining user stories to ensure a structured and methodical development process.

2. Sprint Backlog

As the sole developer on the project, I begin each sprint with a clear planning session where I identify and prioritize the most essential features from the product backlog. These selected tasks are then developed, tested, and finalized by the end of the sprint.

The table under presents the **sprint backlog** for this cycle.

- **Sprint Start Date:** February 17, 2025.
- **Sprint End Date:** March 7, 2025.
- **Work Scale:** One working day (out of Four) equals 7 hours of work (from 10:00 AM to 8:00 PM).
- **Estimated Time:** 84 hours.
- **Sprint Objective:** Add Admin related use cases by implementing user creation, role assignment, profile management, and application configuration, along with user editing and soft deletion capabilities.

Table 10: Sprint Backlog "Management".

ID U.S	User Story	Tasks
1	As a Admin, I should be able to add, modify, and soft delete Users so that I can manage system access securely.	1.1 Prepare the necessary diagrams. 1.2 Develop the use case. 1.3 Test the use case.
2	As a Admin, I want to assign roles to Users so that they have appropriate permissions.	1.1 Prepare the necessary diagrams. 1.2 Develop the use case. 1.3 Test the use case.

Chapter 4. Sprint 2: Management

3	As a Admin I want to impersonate any user (view their dashboard as them) So that I can troubleshoot issues, verify permissions, assist with updates	1.1 Prepare the necessary diagrams. 1.2 Develop the use case. 1.3 Test the use case.
4	As a Admin, I want to add, configure each application (e.g., set commands, update its information) and soft-delete so that the system stays up-to-date.	1.1 Prepare the necessary diagrams. 1.2 Develop the use case. 1.3 Test the use case.
5	As a Admin, I want to define and manage profiles so that permissions can be standardized.	1.1 Prepare the necessary diagrams. 1.2 Develop the use case. 1.3 Test the use case.
6	As an Employee, I want to view my assigned profiles, applications, and request count so that I can track my workload and my given permissions.	1.1 Prepare the necessary diagrams. 1.2 Develop the use case. 1.3 Test the use case.
7	As a Manager, I want to view my employees requests count and information so that I can oversee access approvals.	1.1 Prepare the necessary diagrams. 1.2 Develop the use case. 1.3 Test the use case.
8	As a Admin, I should be able to add, modify, and soft delete Users so that I can manage system access securely.	1.1 Prepare the necessary diagrams. 1.2 Develop the use case. 1.3 Test the use case.

3. Requirements specification

a) Use case diagram

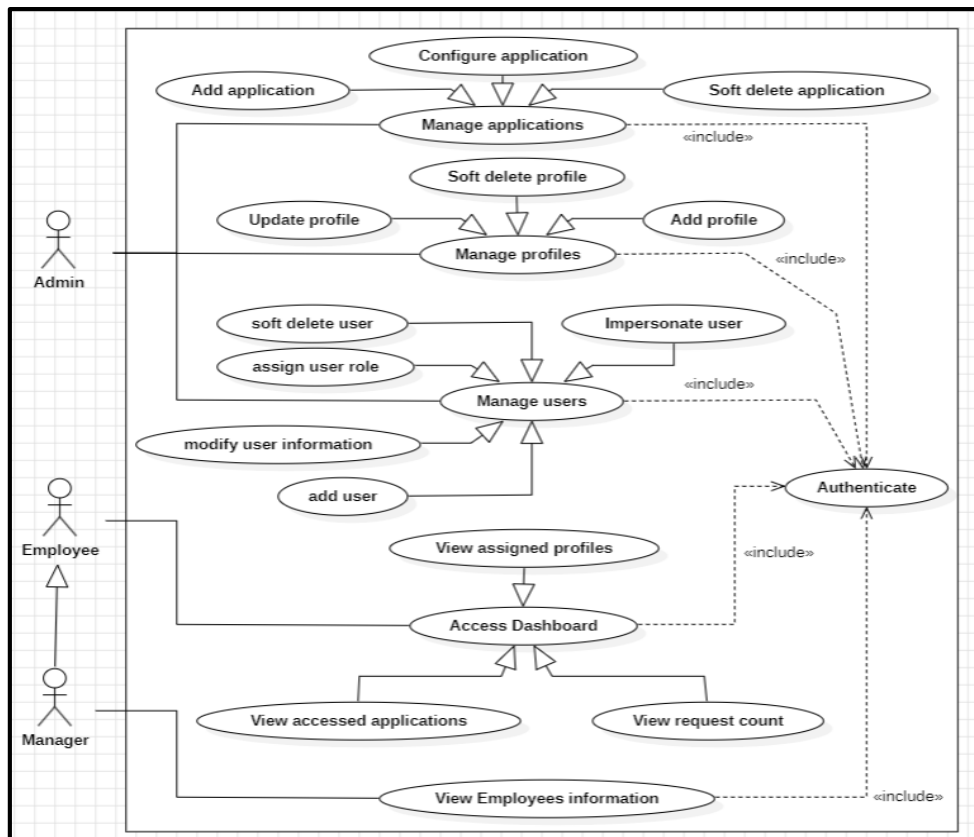


Figure 12: Sprint 2 Use Case Diagram.

Chapter 4. Sprint 2: Management

b) Textual description

The following table, displays a detailed textual description for the sprint “Authentication”

Table 11:Textual Description of UC "Management".

Use case	Management
Actors	User, Admin, Manager, Employee
Objective	Enable Admin to manage apps, profiles, and users; allow Managers to view employee data; and let Employees track their requests
Pre-condition	The user is authenticated and has his specific role
Post condition	• User, app, or profile is created successfully and appears in the relevant list. • In case of duplicates or invalid data, the system displays relevant error messages. • All successful actions are logged. • Managers can see their direct employees' details. • Admins using “Impersonation” feature to see any User’s information.
Nominal Scenario	1. Application Addition: • Admin navigates to "Add App" page. • Admin enters app details (name, IP, linked profiles). • System validates app name uniqueness. • If valid: saves app + profiles, logs action, shows success toast, redirects to app list. 2. Application Configuration: • Admin selects app from list → clicks "Configure." • System loads app and profiles from DB → shows pre-filled configuration page. 3. Soft Delete Application: • Admin initiates deletion → confirms. • System soft-deletes app + profiles, logs action, shows toast, redirects to app list. 4. Profile Addition: • Admin enters profile name and ID. • If unique: profile is created, action logged, toast shown. 5. Profile Update: • Admin edits and submits profile data. • If no conflict: profile is updated, logged, toast shown. 6. Soft Delete Profile: • Admin selects profile → confirms deletion. • System soft-deletes profile and updates list. 7. User Addition: • Admin navigates to "Add User" page. • Admin submits user info (email, role, manager). • If email/ID unique: creates user + manager link, logs action, shows toast, redirects. 8. Impersonate User:

Chapter 4. Sprint 2: Management

	- Admin selects user → system loads user data → redirects to user view. 9. Modify User Information: - Admin edits user details → enters password. - If valid: updates user, logs action, shows toast. 10. Access Dashboard (for Employees): - Employee logs in → sees accessed apps, profiles, and request counts. 11. Access Dashboard (for Managers): - Manager sees all his dashboard data + their employees' requests and info.
Alternative Scenario	1. Duplicate App Name → Error: "Name already exists." 2. Invalid App ID → Error: "App not found." 3. Duplicate Profile Name/ID → Error: "Profile already exists." 4. Profile Update Conflict → Error: "Name/ID exists for this app." 5. Invalid Profile ID (Deletion) → Error: "Profile not found." 6. Duplicate User Email → Error: "Email already registered." 7. Invalid User ID (Impersonation) → Error: "User not found." 8. Incorrect Password (User Edit) → Error: "Incorrect password. Changes discarded." 9. Empty Dashboard → Message: "No data available." 10. Cancel Deletion → Dialog closes; no changes made.

4. Planning

In this section, we'll be exhibiting the **Sequence** diagram as well as the **Class** diagram.

a) Sequence diagram

The following figure, presents the Sequence diagram for this sprint.

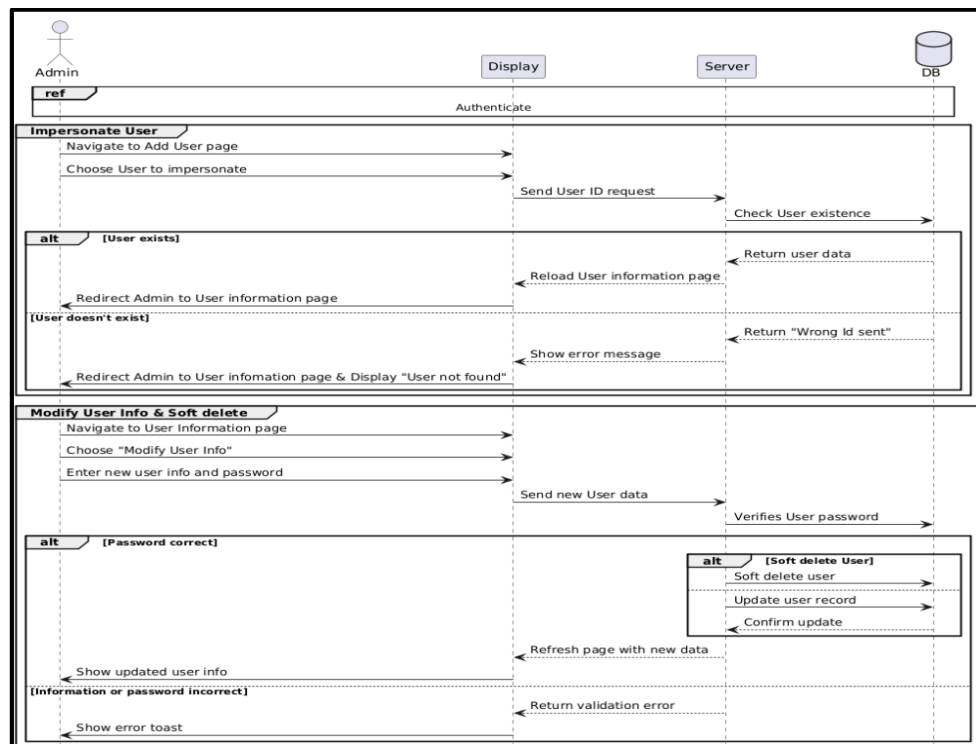


Figure 13: Sequence Diagram for Sprint "Management".

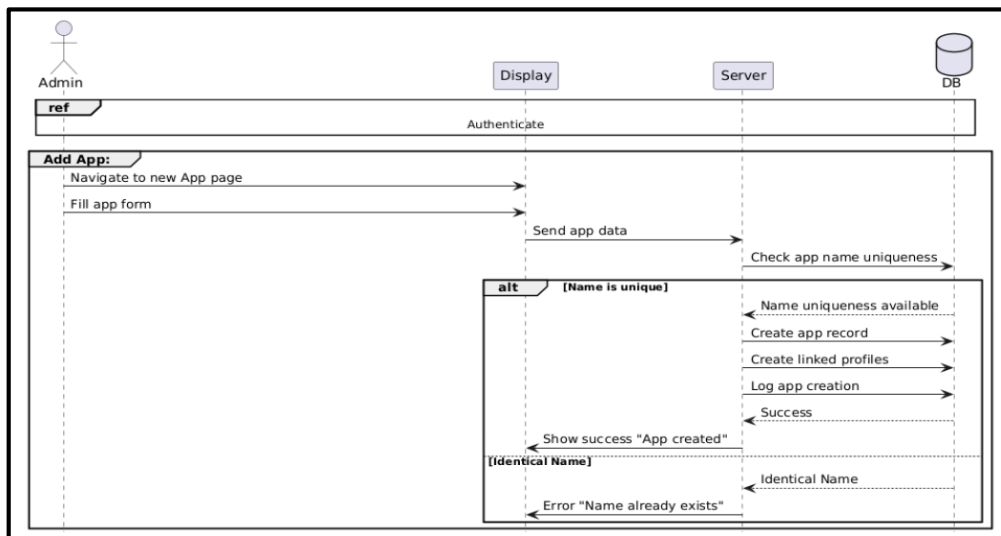


figure 14: Sequence Diagram for Use case "Add Application".

b) Class diagram

The following figure, presents the Class diagram for this sprint.

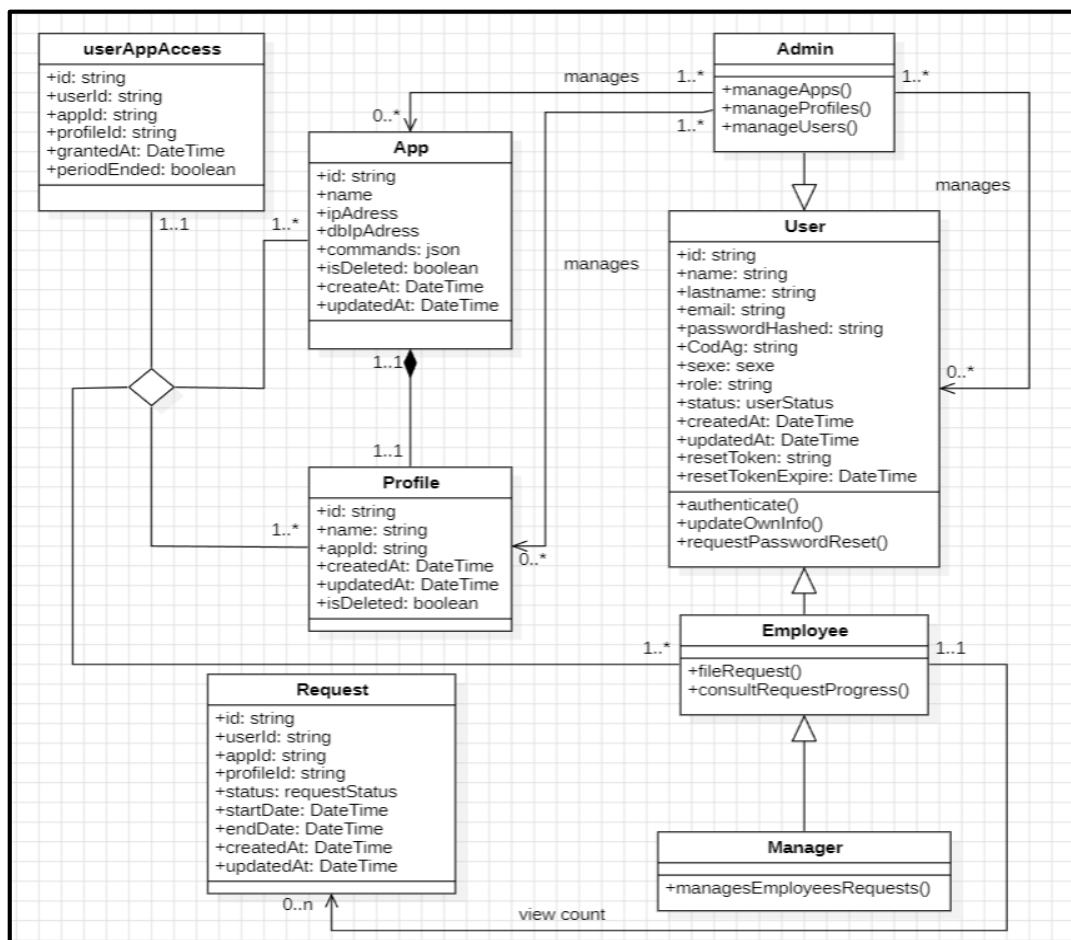


Figure 15: Class Diagram for Sprint "Management".

5. Realization

The following figures show some of the Pages that are present in this sprint, the rest are found in the Annex section.

The screenshot shows the 'Ajouter une application' page. It features a sidebar with navigation options and a main content area with two forms. The 'Nouvelle application' form has fields for application name, IP addresses, and SQL commands. The 'Ajouter des Profils' form has fields for profile name and ID. The bottom of the page displays the user's name and email.

Figure 16: Application Addition Page.

The screenshot shows the 'Tableau de bord Manager' page. It features a sidebar with navigation options and a main content area with several sections. The 'Informations Personnelles' section shows the user's name, email, and role. The 'Mes Demandes' section shows the total number of requests and their status. The 'Employés Gérés' section shows a table of employees. The 'Demandes Employés' section shows the total number of requests and their status.

Nom	Prénom	Email	Rôle
Amir	Smati	john.doe@example.com	Employé
Alice	Doe	alice@example.com	Manager

Figure 17: Manager Dashboard.

6. Conclusion

After several meetings and revisions, our supervisor validated this sprint's functionalities, which concentrated on admin features including creating and managing profiles, users, and applications, as well as modifying them enabling us to progress confidently to the next sprint.

Chapter 5

Sprint 3: Request handling (Request Management, Request Execution)

1. Introduction

In this chapter, we planned the sprint by defining user stories to ensure a structured and methodical development process.

2. Sprint Backlog

As the sole developer on the project, I begin each sprint with a clear planning session where I identify and prioritize the most essential features from the product backlog. These selected tasks are then developed, tested, and finalized by the end of the sprint.

The table under presents the **sprint backlog** for this cycle.

- **Sprint Start Date:** March 10th, 2025.
- **Sprint End Date:** March 29th, 2025.
- **Work Scale:** One working day (out of Four) equals 7 hours of work (from 10:00 AM to 8:00 PM).
- **Estimated Time:** 84 hours.
- **Sprint Objective:** Implement Request filing, progression, escalating and monitoring for all Users.

3. Requirements specification

a) Use case diagram

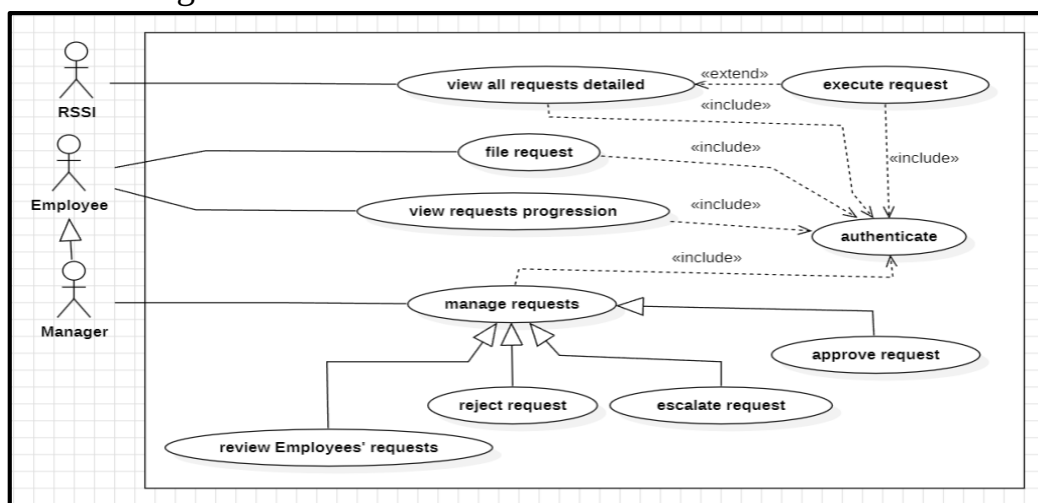


Figure 18: Sprint 2 Use Case Diagram.

Chapter 5. Sprint 3: Request handling

b) Textual description

Table 12: Textual Description of UC "Request Handling".

Use case	Management
Actors	RSSI, Manager, Employee
Objective	Enable users to submit access requests, allow managers to approve/reject/escalate them, and authorize access execution by RSSI.
Pre-condition	User is authenticated
Post condition	- Access is granted if request is approved and executed - All key actions are logged and notifications sent - Invalid or conflicting requests are rejected with appropriate feedback
Nominal Scenario	- Submit access request: - Employee navigates to the request page and fills in the form. - System validates input and ensures no duplicate pending or conflicting approved requests exist. - If valid: creates request, assigns to manager, logs action, emails manager, shows toast: <i>"Request submitted successfully."</i> - View & Review Requests: - Manager accesses their pending approvals. - System fetches relevant requests and displays them. - Approve, Reject, or Escalate: - Manager selects a request and chooses an action: Accept: Marks approval as treated; if all managers approve, request is marked as APPROVED and RSSI is notified. Reject: Request status is updated to REJECTED; employee is notified. Escalate: Request is transferred to a higher-level manager; both old and new managers are logged and notified. - Execute Request (RSSI): - RSSI views the full request list and selects a request to execute. - System shows request details and approval history. - If the request is in APPROVED status, the "Execute" button becomes active. - Upon execution: - Access scripts are generated. - Scripts are stored and access records created in the DB. - The request is updated to EXECUTED. - Action logged. - Notify request submitter (Employee/Manager). - Track Request Status: - Employee/Manager navigates to their request history page. - System displays approval steps and status updates.
Alternative Scenario	Duplicate pending request: Error toast shows "Pending request exists." Conflicting approved access: System warns about overlapping active access. No requests filed: System shows empty table. No assigned manager: System blocks request creation. Execution failure: Error toast shows generation issue. Request not in APPROVED state: Execute button disabled. Manager not found (on escalation): Escalation is blocked and logged. Missing fields: Toasts shown to address this issue

4. Planning

In this section, we'll be exhibiting the **Sequence** diagram as well as the **Class** diagram that was used to achieve this Sprint.

a) Sequence diagram

The following figure, presents the Sequence diagram for this sprint.

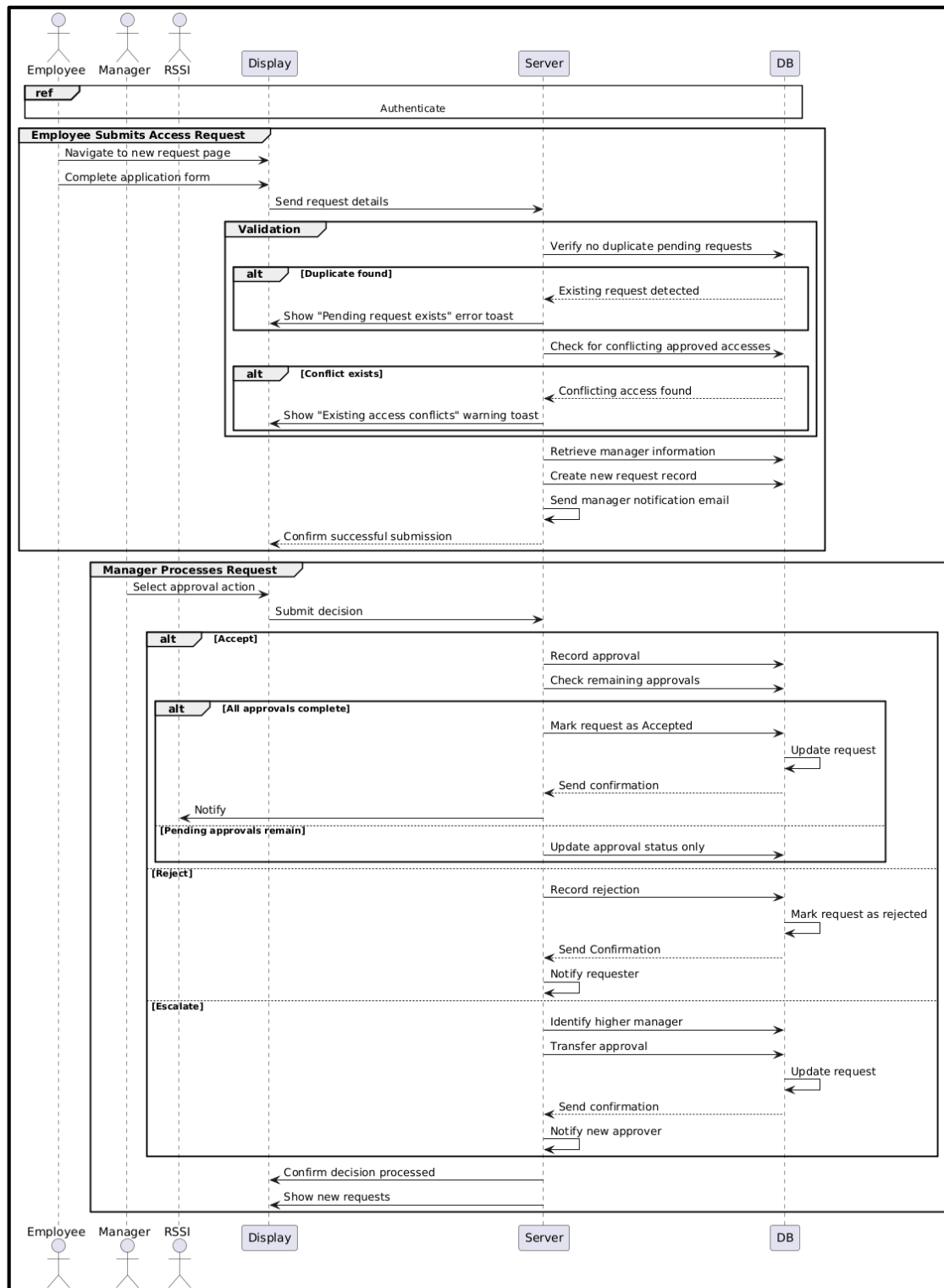


Figure 19: Sequence Diagram for "Request Handling" Sprint.

b) Class diagram

The following figure, presents the Class diagram for this sprint.

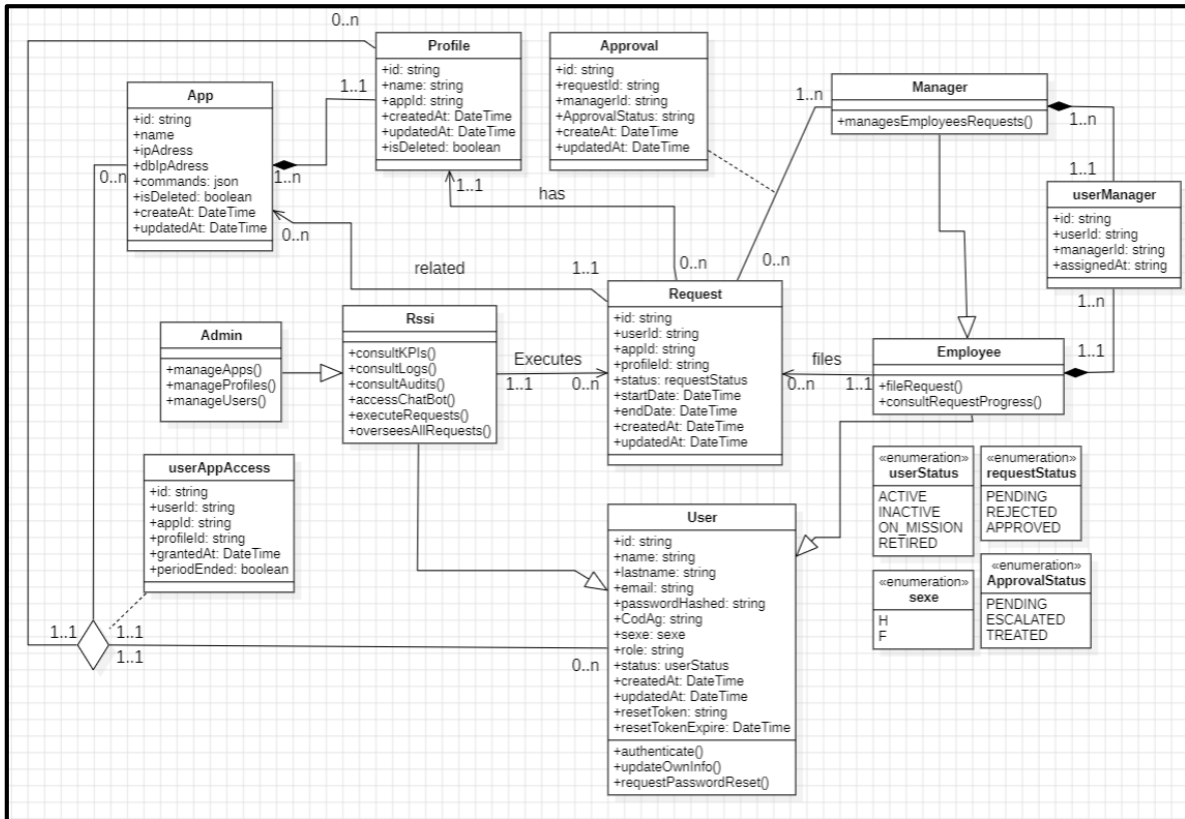


Figure 20: Class Diagram for "Request Handling" Sprint.

5. Realization

The following figures show some of the Pages that are present in this sprint, the rest are found in the Annex section.

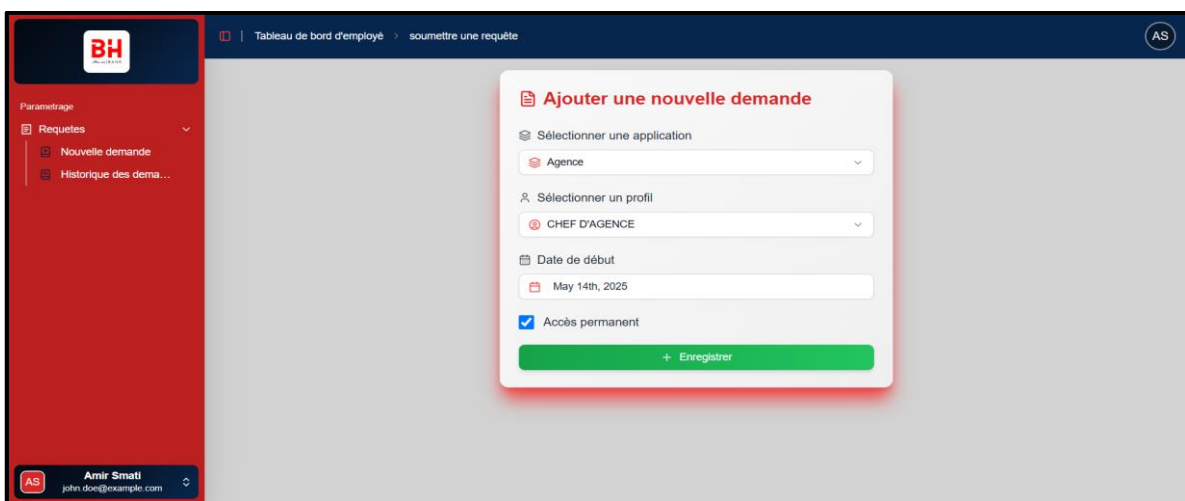


Figure 21: Filing Request Page.

Chapter 5. Sprint 3: Request handling

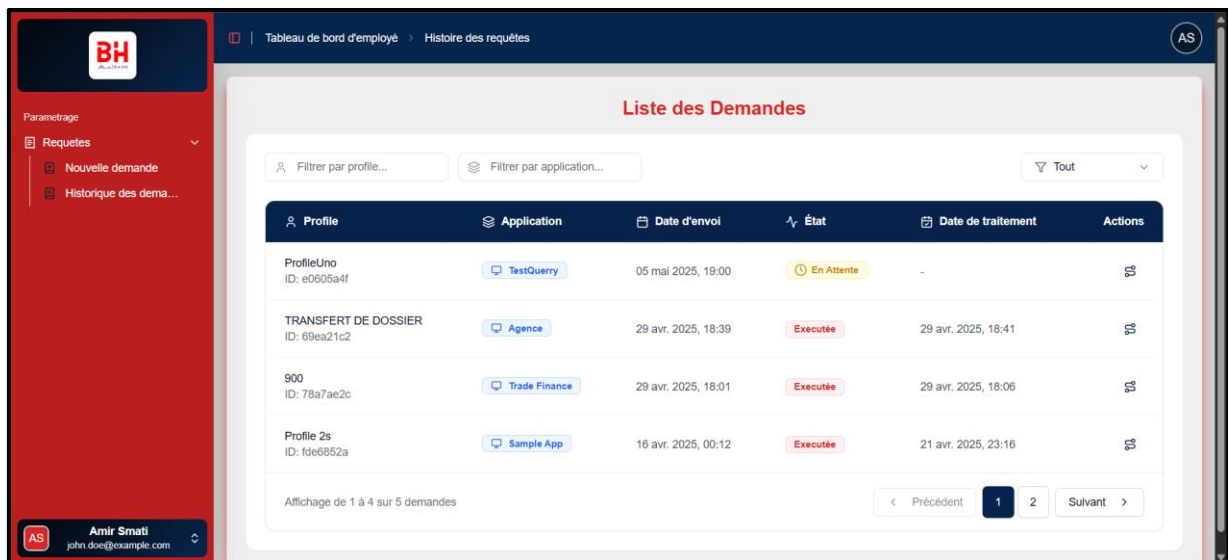


Figure 22: Employee Request Page.

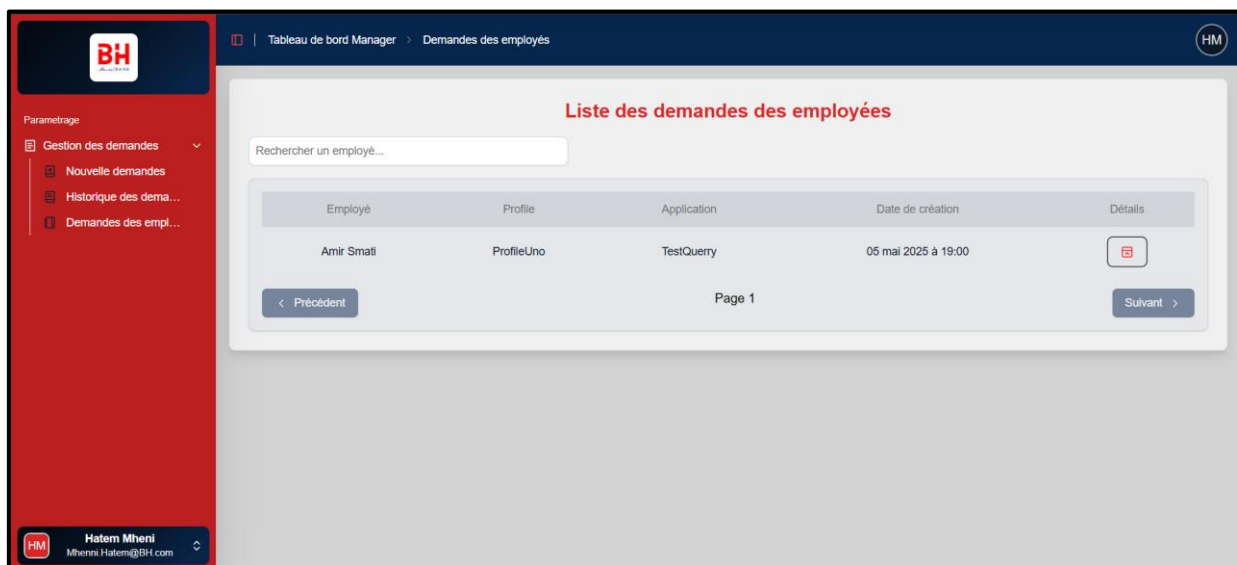


Figure 23: Employees Requests Page.

6. Conclusion

After several meetings and adjustments, our supervisor validated this sprint's functionalities, which focused on request handling including how requests are filed, notifying managers, implementing escalation logic, and managing app access allowing us to confidently move forward to the next sprint.

Chapter 6

Sprint 4: Control (Logs & Audit trails consulting, KPIs consulting, Auditron implementation)

1. Introduction

In this chapter, we planned the sprint by defining user stories to ensure a structured and methodical development process.

2. Sprint Backlog

As the sole developer on the project, I begin each sprint with a clear planning session where I identify and prioritize the most essential features from the product backlog. These selected tasks are then developed, tested, and finalized by the end of the sprint.

The table under presents the **sprint backlog** for this cycle.

- **Sprint Start Date:** April 1st, 2025.
- **Sprint End Date:** Mai 3rd, 2025.
- **Work Scale:** One working day (out of Four) equals 5 hours of work (from 10:00 AM to 5:00 PM).
- **Estimated Time:** 100 hours.
- **Sprint Objective:** Implement Request filing, progression, escalating and monitoring for all Users.

3. Requirements specification

a) Model Context Architecture (MCP/MCO Integration)

In this sprint, we implemented a layered architecture centered around a custom-designed **Model Context Protocol (MCP)** and a dedicated **Model-Context Operator (MCO)** that together form the foundation of the Auditron assistant. This architecture is a significant step forward from traditional chatbot or rule-based systems. It introduces a **context-sensitive, role-aware intelligence layer** that governs every user interaction.

Chapter 6. Sprint 4: Control

The **MCP** acts as the core logic controller, responsible for interpreting natural language input by classifying it (general query, or audit task), extracting intent and parameters, and verifying permissions based on user roles such as Admin or RSSI. The **MCO**, Auditoron, then executes the appropriate actions, whether it's answering conversational questions, automating secure database tasks, or summarizing uploaded audit log files while maintaining strict adherence to contextual rules and operational boundaries.

This layered setup is not just a technical enhancement; it represents a **paradigm shift** in how users interact with critical banking systems. It transforms static interfaces and fragmented dashboards into a **unified, intelligent conversational gateway**. Through natural language alone, authorized users can perform complex, high-traceability operations, all without compromising **security, compliance, or auditability**.

The result is a prototype showing how **AI can be safely integrated** into financial systems. It sets the stage for future expansion into **multi-database setups, predictive analysis, and autonomous audits**. Auditoron goes beyond responding it **understands, acts, and safeguards**. It reflects the future of secure, intelligent digital banking.

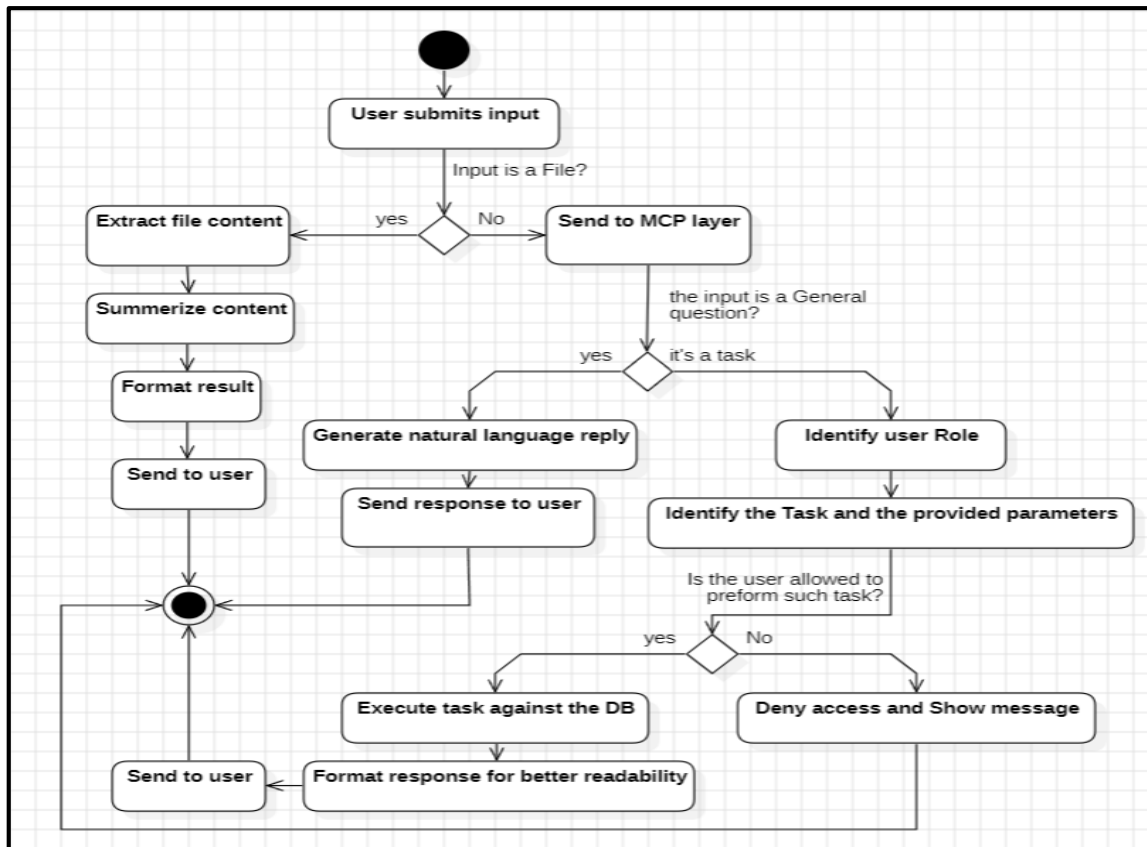


Figure 24:AudiTron Architecture.

b) Use case diagram

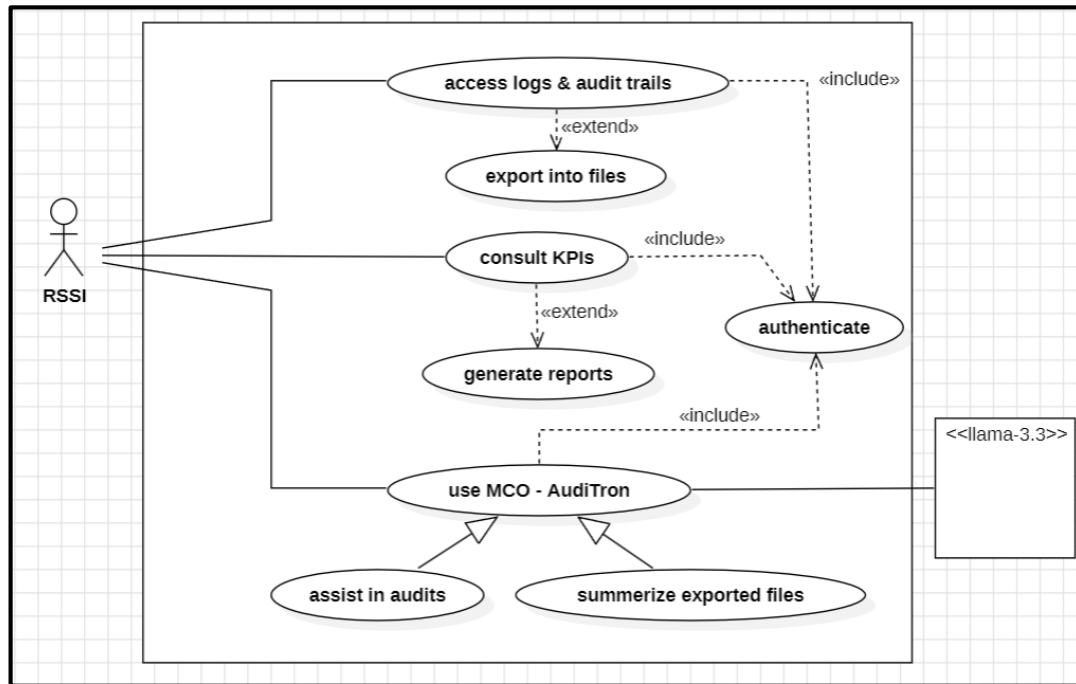


Figure 25:Sprint 4 Use Case diagram.

c) Textual Description

The following figure, presents the Class diagram for this sprint.

Table 13:Textual Description of UC "Control".

Use case	Control
Actors	Admin, RSSI
Objective	Enable Admins and RSSIs to perform intelligent audit-related operations using the Auditron assistant (MCO), including consulting logs, KPIs, summarizing exported files, and executing audit-specific queries through a role-aware, context-driven interface.
Pre-condition	- User is authenticated - Users' role must be either RSSI or Admin
Post condition	- Queries return accurate responses or formatted data. - Documents are processed and summarized. - Logs/KPIs are filtered, displayed, or exported as requested.
Nominal Scenario	- General AudiTron Conversation: - RSSI navigates to AudiTron page. - Asks a general question through the interface. - System (via MCP): - Classifies question type. - Sends contextual prompt. - Displays formatted natural language response. - Audit Data Request: - RSSI submits a database -related audit question. - System (via MCP):

	(1) Sends structured prompt to the LLM. (2) Uses classifier to identify task intent. (3) Validates permissions and parameters. (4) Queries database using relevant Prisma function. (5) Receives JSON response. (6) Applies structured formatting and returns a readable answer. (7) Displays response. 3. Document processing: • RSSI uploads a CSV/export file. • System: (1) Extracts and formats content. (2) Sends to LLM for summarization. (3) Displays structured summary. 4. Fetch Logs/Audits/KPIs: • RSSI applies filters on logs/KPIs page. • System: (1) Fetches matching data from database. (2) Displays results. 5. Export Logs/Audits/KPIs: • RSSI selects logs/KPIs to export. • System: (1) Formats data for printing/download (2) Generates downloadable file.
Alternative Scenario	• General Conversation Failure: Error toast shows "Pending request exists." • Conflicting approved access: System warns about overlapping active access. • No requests filed: System shows empty table. • No assigned manager: System blocks request creation. • Execution failure: Error toast shows generation issue. • Request not in APPROVED state: Execute button disabled. • Manager not found (on escalation): Escalation is blocked and logged. • Missing fields: Toasts shown to address this issue

4. Planning

In this section, we'll be exhibiting the **Sequence** diagram as well as the **Class** diagram that was used to achieve this Sprint.

a) Sequence diagram

The following figure, presents the Sequence diagram for this sprint.

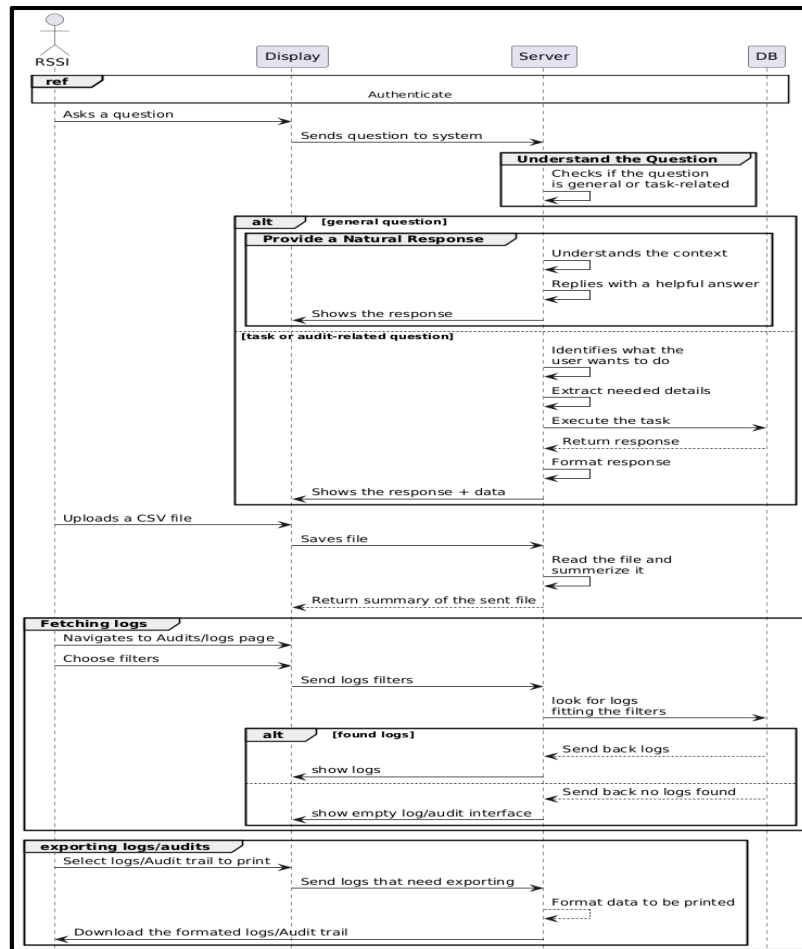


Figure 26: Sequence Diagram for Sprint 4.

b) Class diagram

The following figure, presents the Class diagram for this sprint.

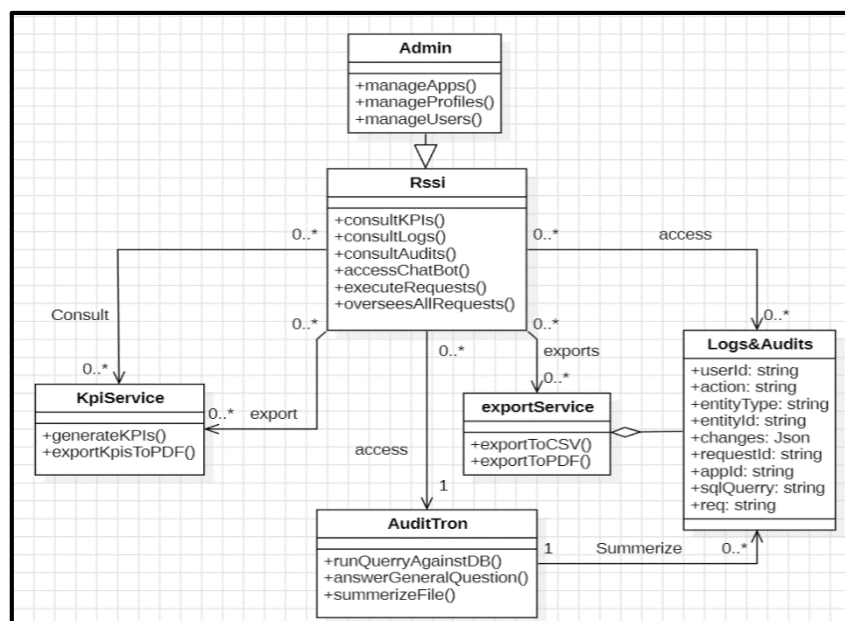


Figure 27: Class Diagram for Sprint "Control".

4. Realization

The following figures show some of the Pages that are present in this sprint, the rest are found in the Annex section.

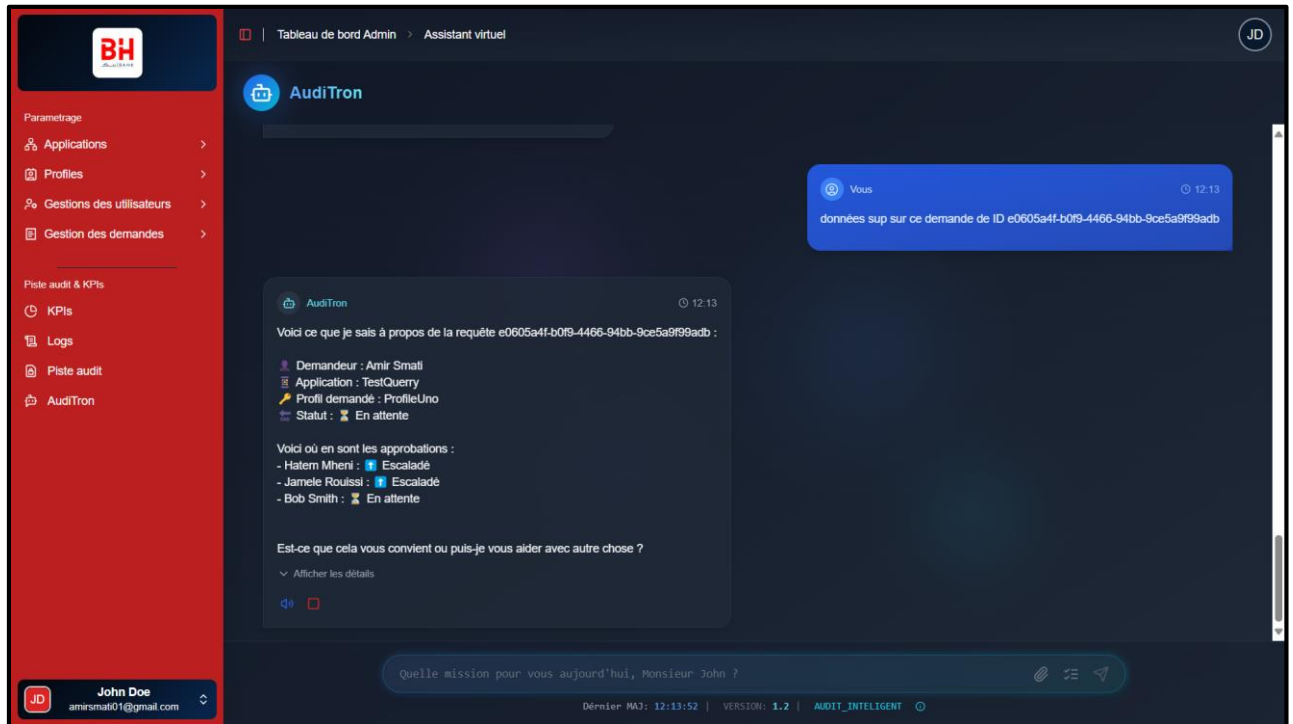


Figure 28: AudiTron Page.

5. Conclusion

The final sprint focused on building robust KPI tracking, comprehensive logging, and enhanced auditing features. This phase also saw the successful integration of an AI assistant that naturally evolved into a Hybrid Model Context Protocol (MCP) system providing secure, context-driven, and intelligent audit support. Together, these advancements significantly strengthen the project's capabilities in access management and compliance oversight.

General Conclusion

SecureAccess is a forward-looking and strategic response to long-standing challenges in identity and access management (IAM) within banking a domain where security, compliance, and operational efficiency are critical. Unlike legacy IAM platforms, SecureAccess was purpose-built to bridge fundamental gaps such as outdated access models, non-intuitive workflows, and the complete absence of intelligent operational agents.

A defining innovation of SecureAccess is the integration of a custom-built **Model Context Protocol (MCP)** powering **Auditron**, an intelligent operational assistant acting as a **Model-Context Operator (MCO)**. This architecture represents a breakthrough in audit digitalization and intelligent access handling. Auditron does more than respond to queries it understands intent, validates roles, and executes compliant actions through natural language, while ensuring enterprise-grade traceability and control.

Rather than relying on any third-party MCP/MCO framework, this protocol-driven system was designed and implemented from the ground up. The result is not just a technical feature it's a **paradigm shift** in how financial institutions can interface with complex systems: through human-like dialogue, securely governed by structured context, permissions, and real-time auditability. By acting as a secure conversational gateway, Auditron empowers even non-technical users to carry out powerful operations within strict compliance boundaries.

This project marks a concrete **stepping stone toward the future of AI-driven governance** in banking. The MCP-based interaction model supports intelligent task routing, dynamic access enforcement, and contextual audit logging all while remaining transparent, secure, and scalable. As banks face increasing complexity, the long-term potential for expanding this architecture into areas like predictive analytics, anomaly detection, and autonomous access governance becomes both feasible and inevitable.

In addition to the MCP/MCO layer, SecureAccess incorporates modern technologies such as Next.js and Prisma, combined with intuitive user interface design and robust KPI visualization. Each component was developed through iterative, agile methods with a focus on operational realism and security.

In conclusion, SecureAccess does more than meet current industry standards it **reimagines the future of IAM in banking**. It sets a new benchmark for secure, adaptive, and intelligent systems and lays the groundwork for broader AI-powered transformation across financial technology.

Webography

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Annex

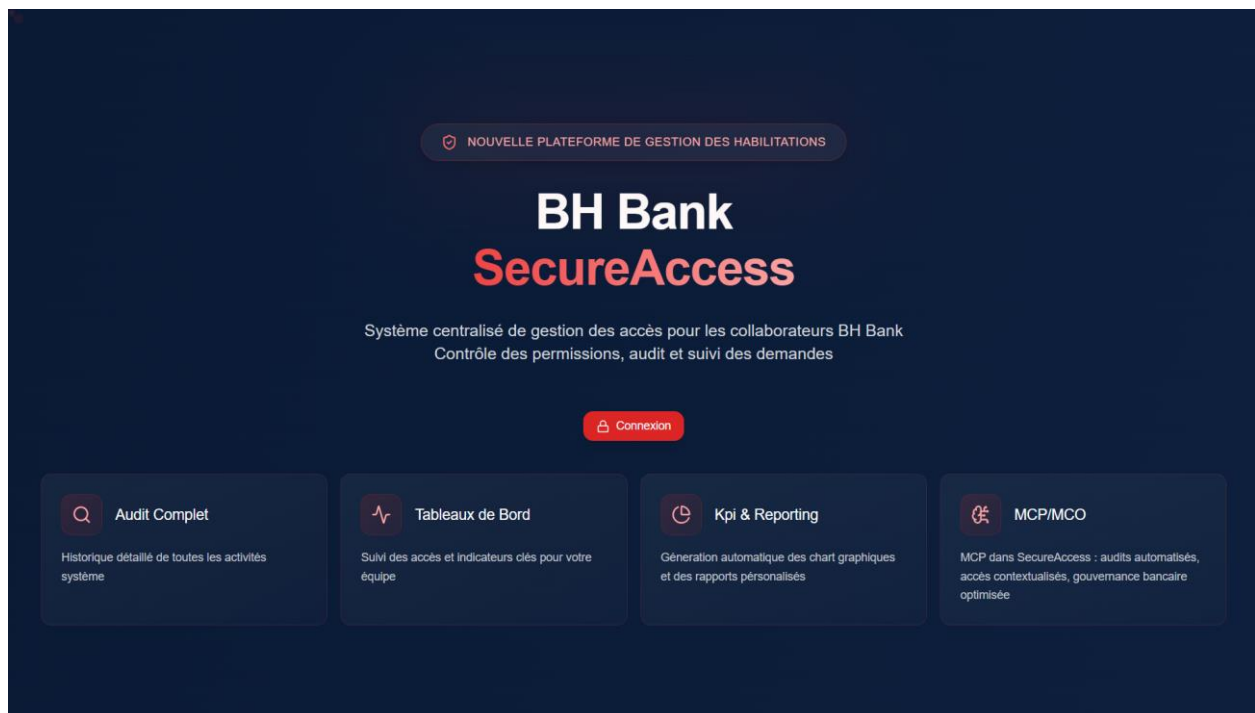


Figure 29: Hero Section of the Landing page.



Figure 30: Why Section of the Landing page.

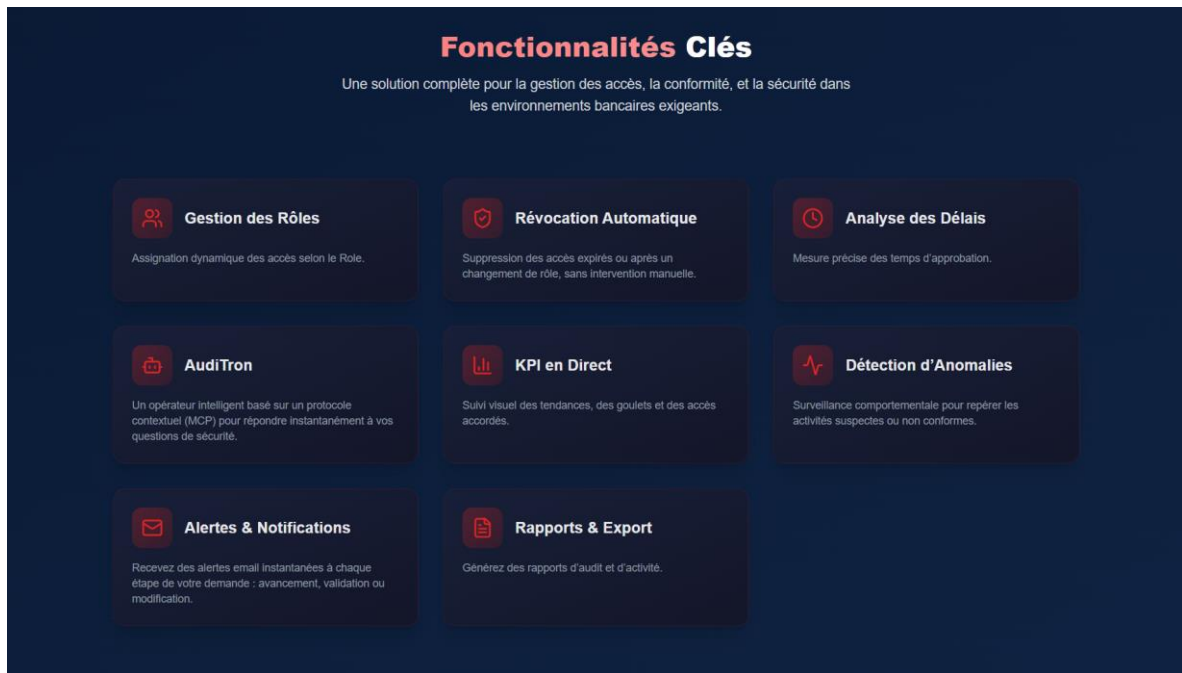


Figure 31: Key Functions Section of the Landing page.

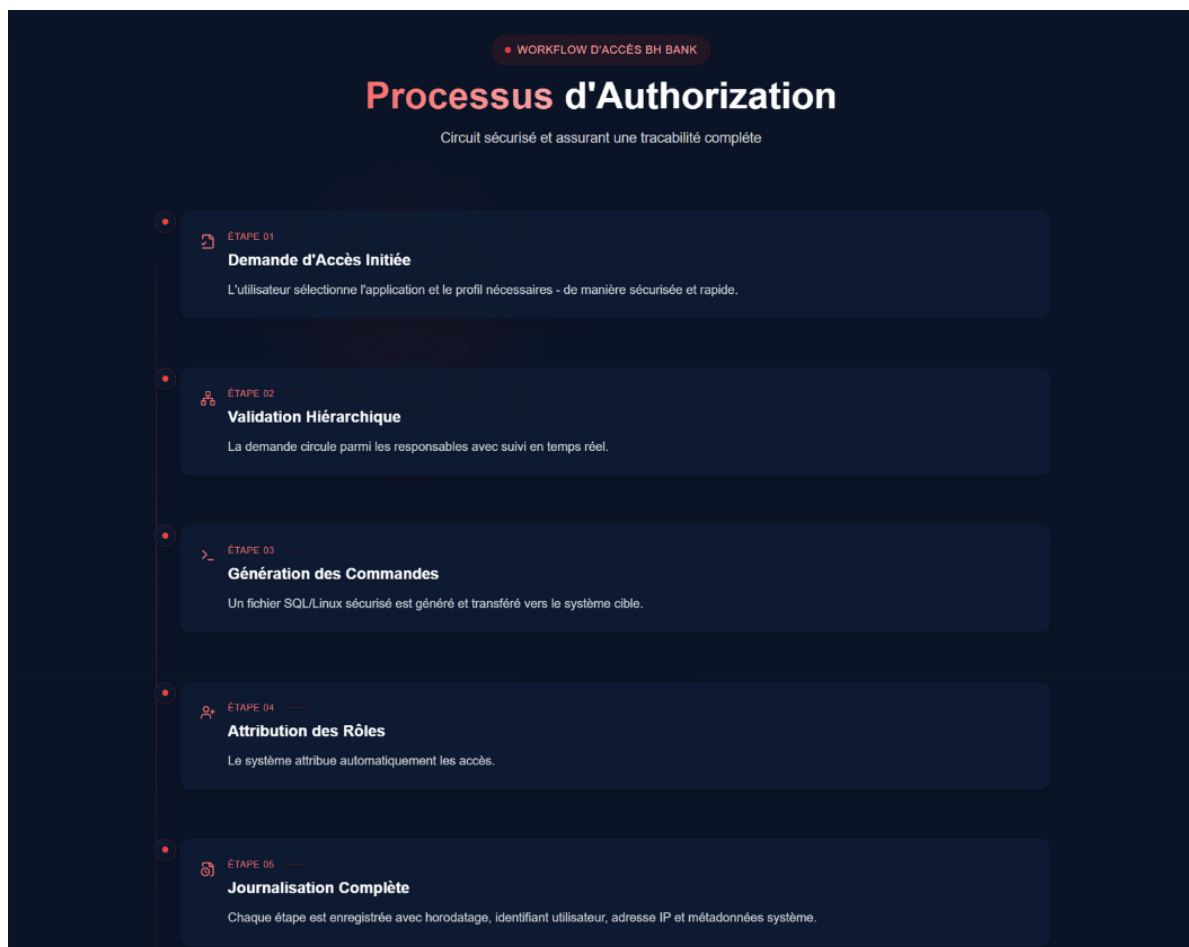


Figure 32: Process Section of the Landing page.

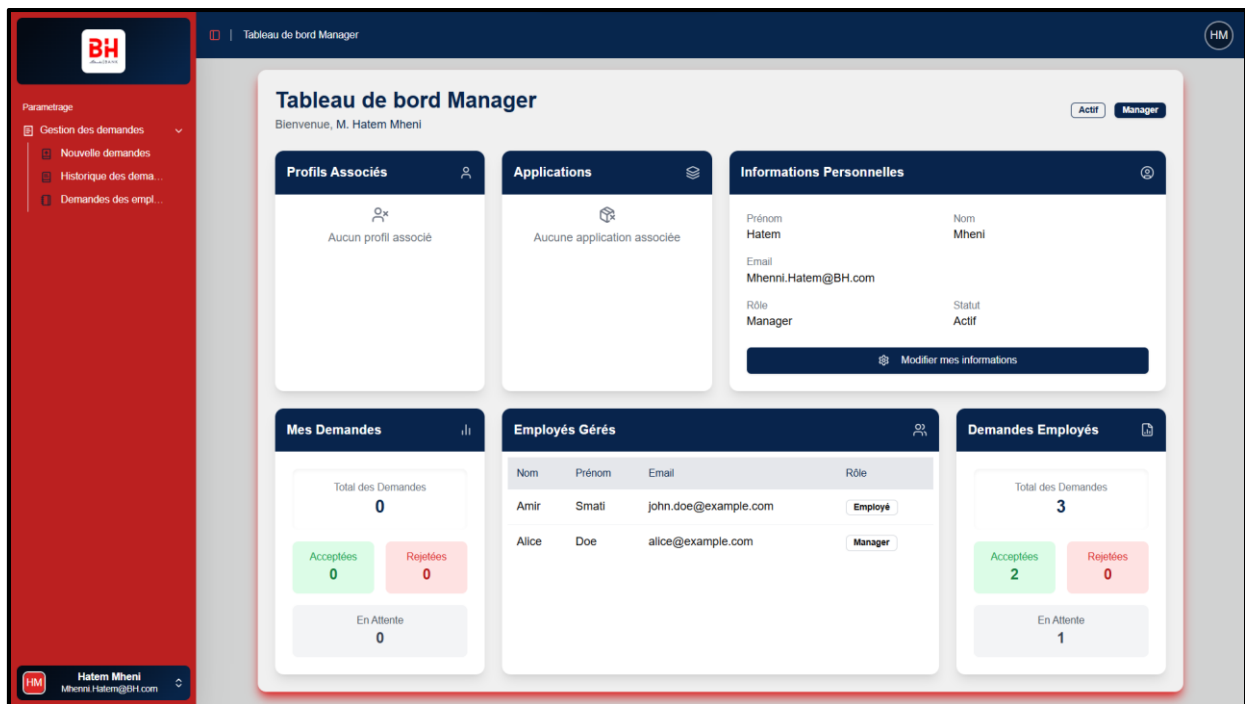


Figure 33: Managers' Dashboard.

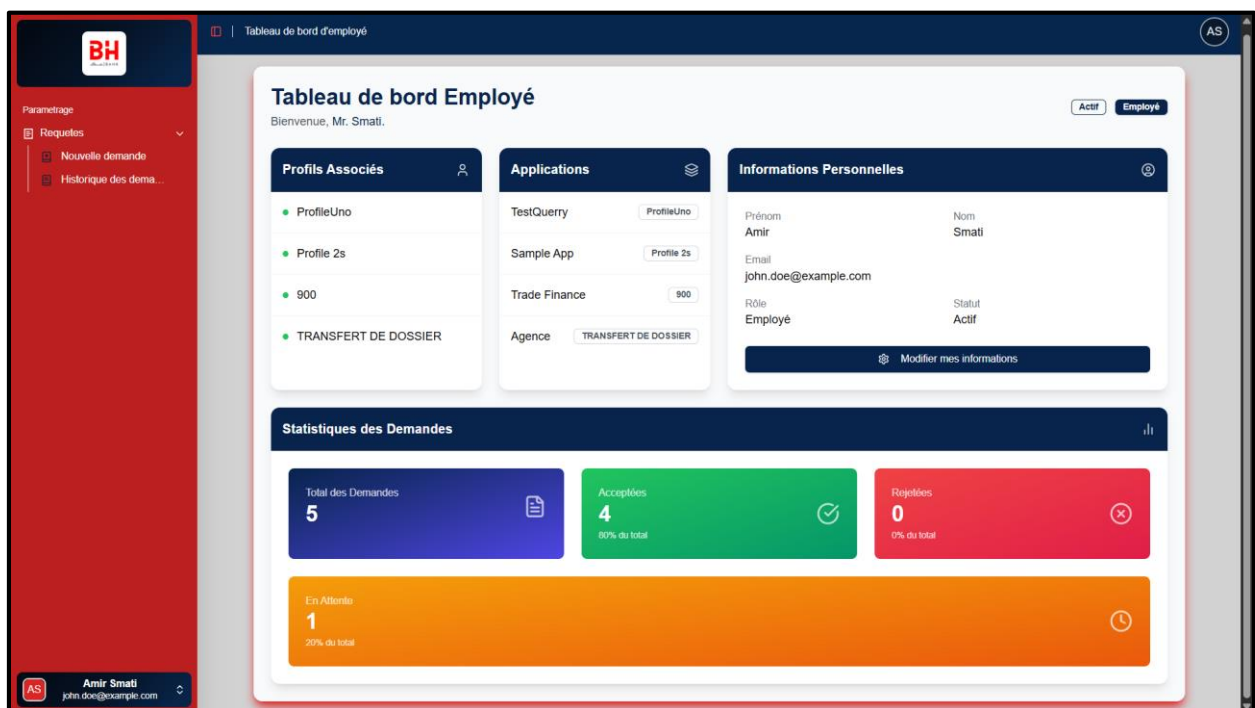


Figure 34: Employees' Dashboard

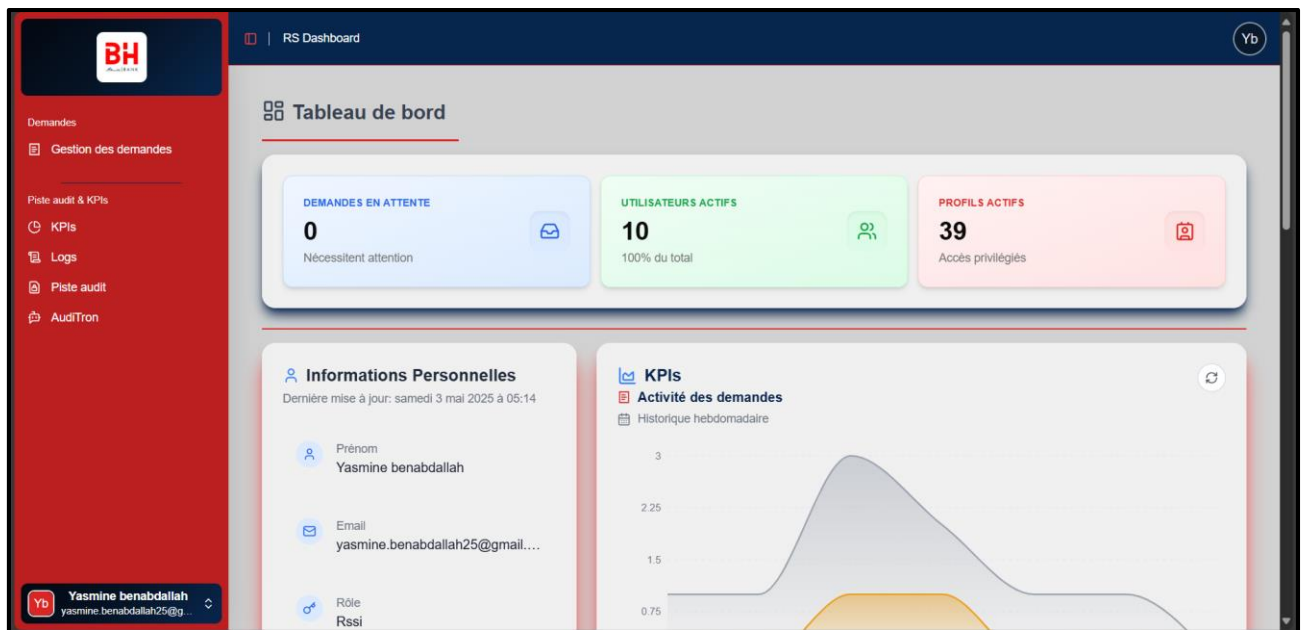


Figure 35:RSSIs' Dashboard.

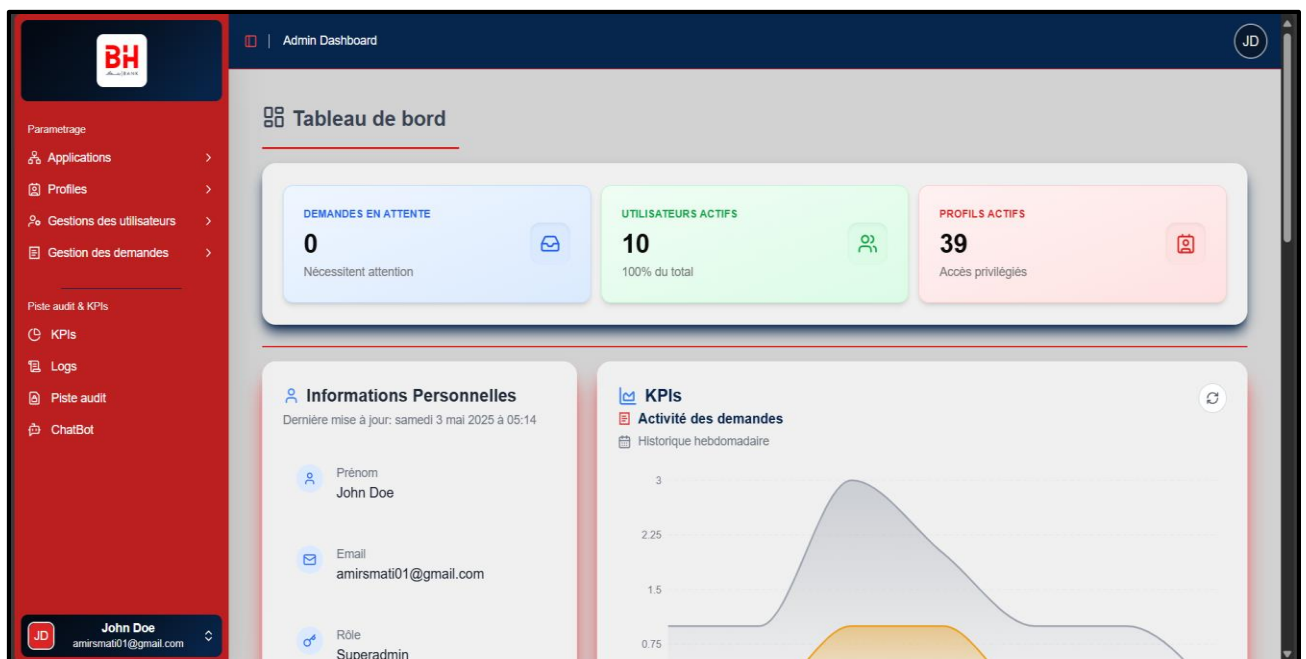


Figure 36: Admins' Dashboard.

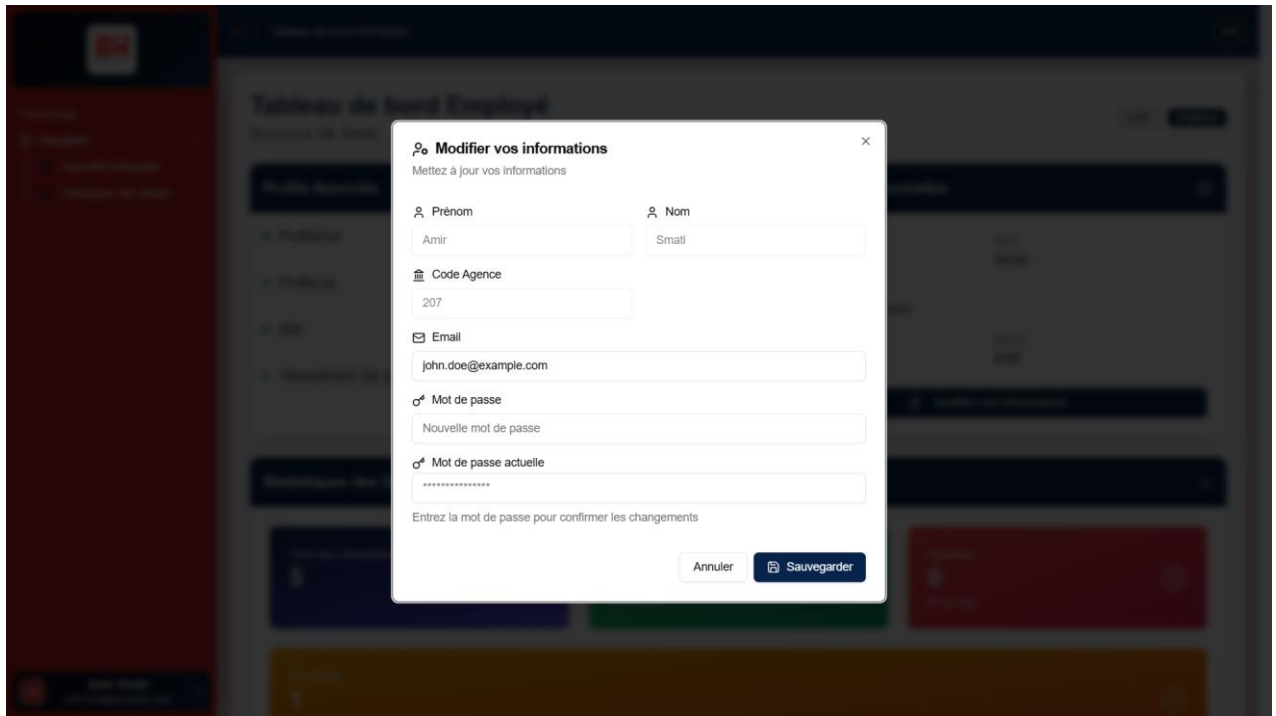


Figure 37: Information Update Component.



Figure 38: Password Reset page.

Réinitialisation de mot de passe

Bonjour,

Une demande de réinitialisation de mot de passe a été effectuée pour votre compte. Cliquez sur le bouton ci-dessous pour continuer :

Réinitialiser le mot de passe

Important : Ce lien expirera automatiquement dans 24 heures. Si vous n'êtes pas à l'origine de cette demande, veuillez ignorer cet email.

Si le bouton ne fonctionne pas, copiez-collez ce lien dans votre navigateur :

<http://localhost:3000/reset-password?token=88d7e2359aa40ba2bca9282b2e8d3c8223c2251e41462e6db8b663fb1a935050>

Pour toute question, contactez notre équipe à support@bhbank.com

© BH Bank - Service client

Figure 39: Password Reset Mail.

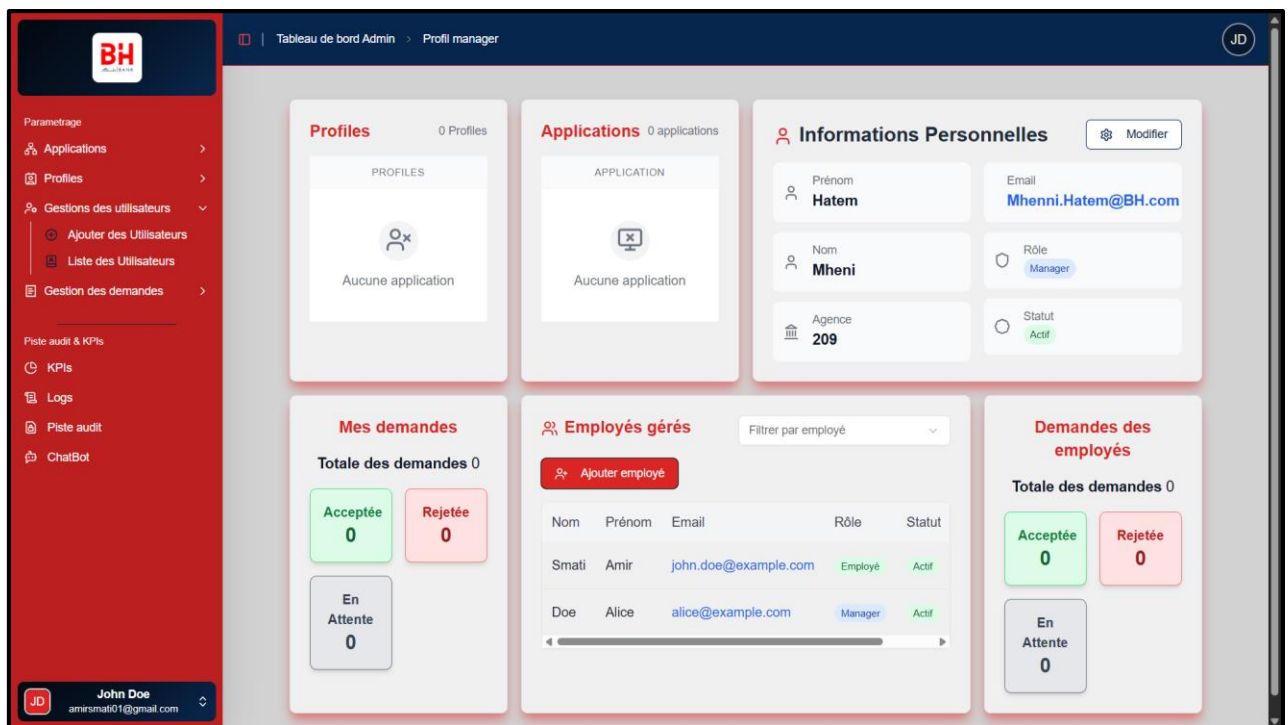


Figure 40: Impersonation Feature Preview (Manager).

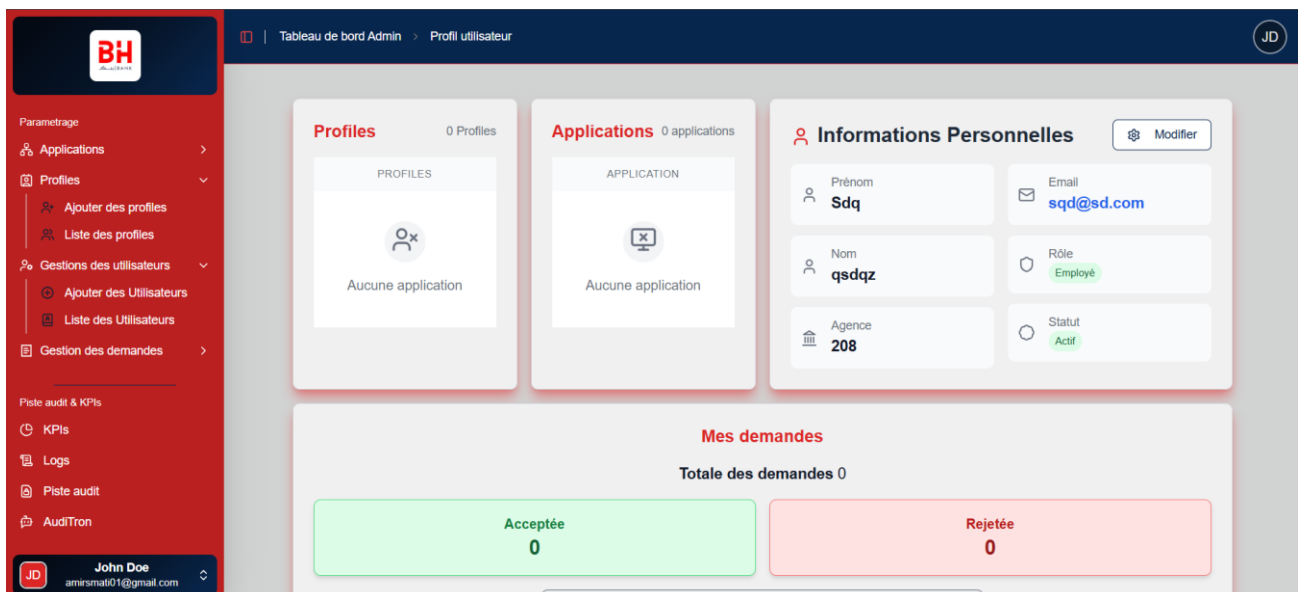


Figure 41: Impersonation feature preview (Employee).

Modifier le Profil

Mettez à jour les informations de l'utilisateur

Id: 11

Prénom: Hatem

Nom: Mheni

Email: Mhenni.Hatem@BH.com

Code Agence: 209

Rôle: Manager

Statut: Actif

Mot de passe: Mot de passe de l'utilisateur

Entrez la mot de passe pour confirmer les changements

Annuler Sauvegarder

Figure 42: Modify User's Information (Admin Perspective).

Ajouter un utilisateur

Nouvel Utilisateur

Code agence: Code Agence

ID: Identifiant unique

Prénom: Prénom

Nom: Nom de famille

Email: email@exemple.com

Mot de passe: *****

Sexe: Homme

Rôle: Utilisateur

Rechercher un Manager: Rechercher un manager...

Manager assigné: Sélectionner un manager

Ajouter l'utilisateur

Figure 43: User Addition Page.

JD

Tableau de bord Admin > Configuration d'application

← Configuration de Megara

Megara

📍 IP Application:
192.0.0.4

📌 Statut:
● Active
📅 Créé le:
26/02/2025

👤 4
Profils

> 1
Commandes

📄 Non connectée

🔍 Profils d'Application +

👤 2
Créé le 26/02/2025

👤 3
Créé le 26/02/2025

👤 A1
Créé le 10/03/2025

👤 1 Désactivée
Créé le 26/02/2025

>_ Commandes d'Application +

Clé
add

Type
SQL

Requête
add \${userid}

⚠ Zone de Danger

Désactiver l'Application
Cela désactivera l'application et tous ses profils associés

🚫 Désactiver

📄 Enregistrer les modifications

Figure 44: Application Configuration Page.

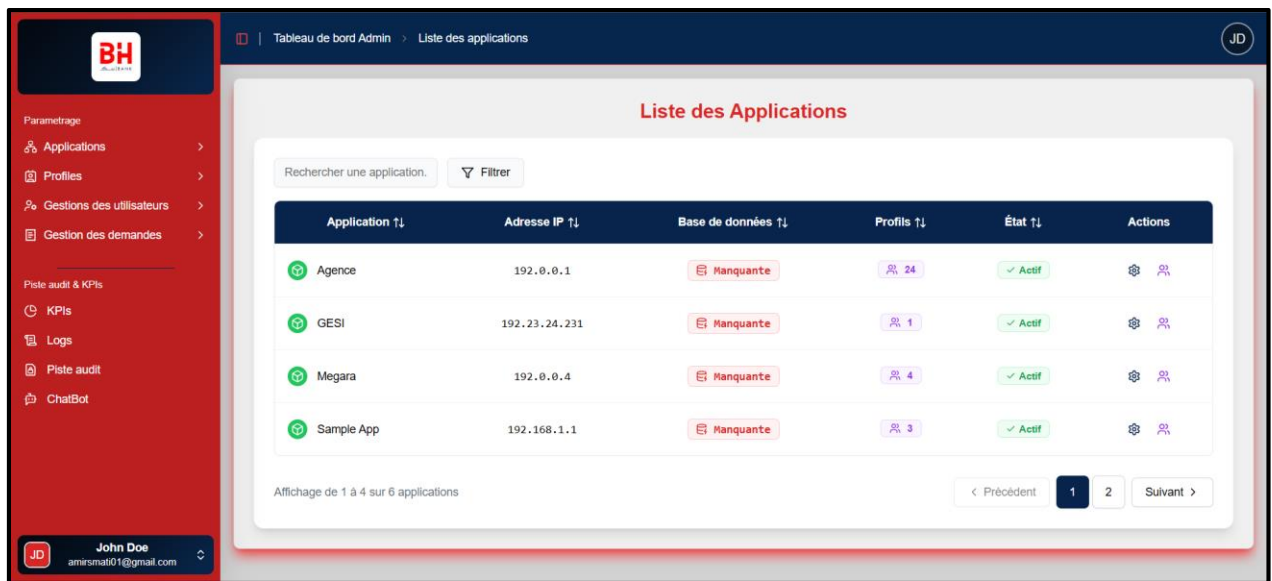


Figure 45: Application List Page.

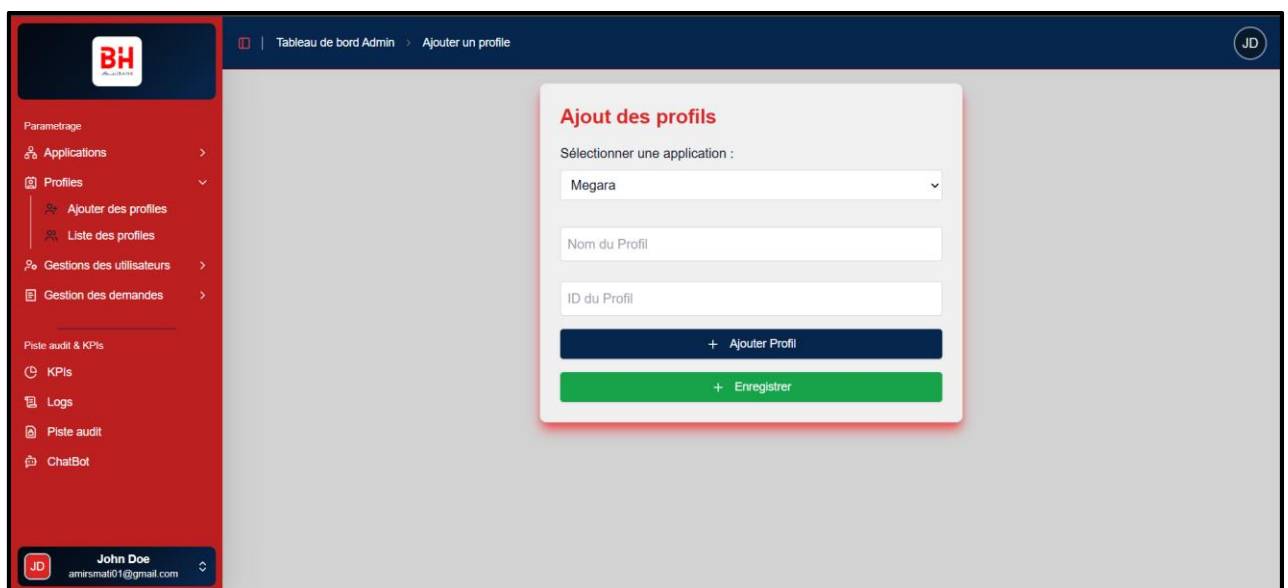


Figure 46: Profile Addition Page.

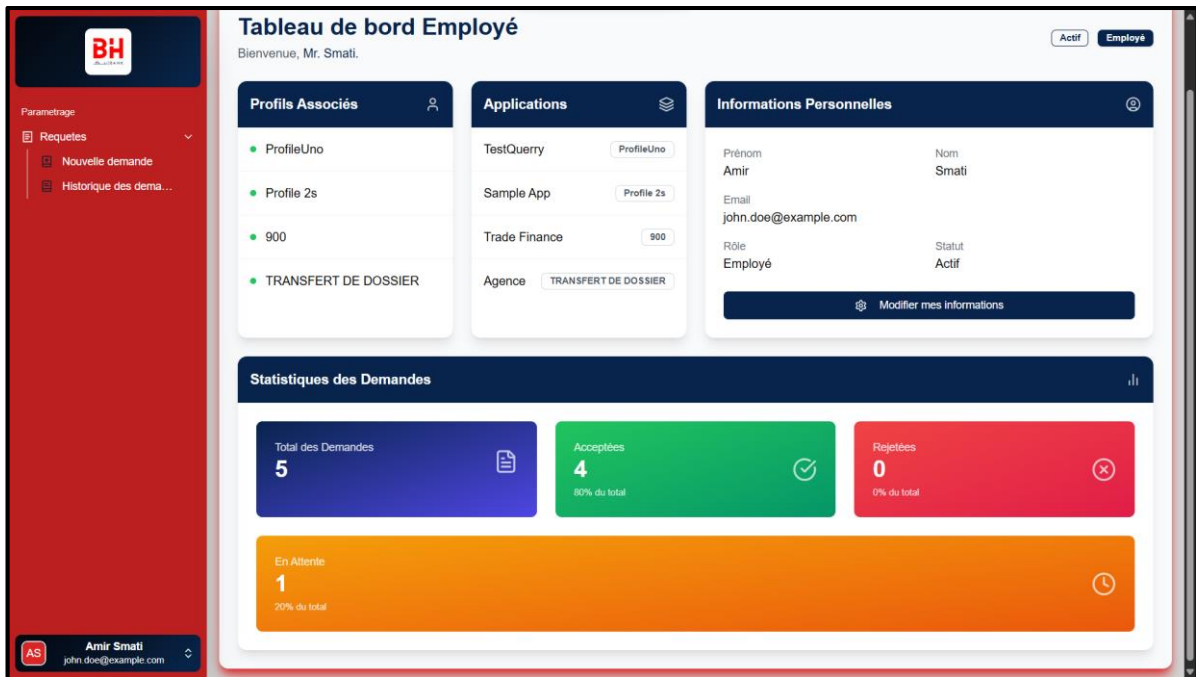


Figure 47: Employee Dashboard.

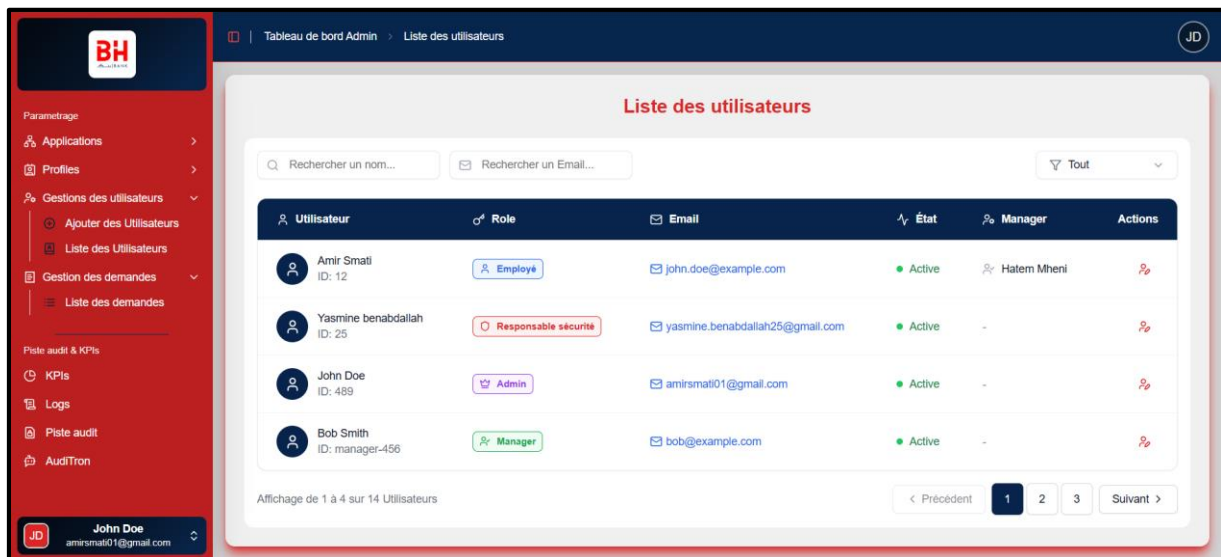


Figure 48: User List Page.

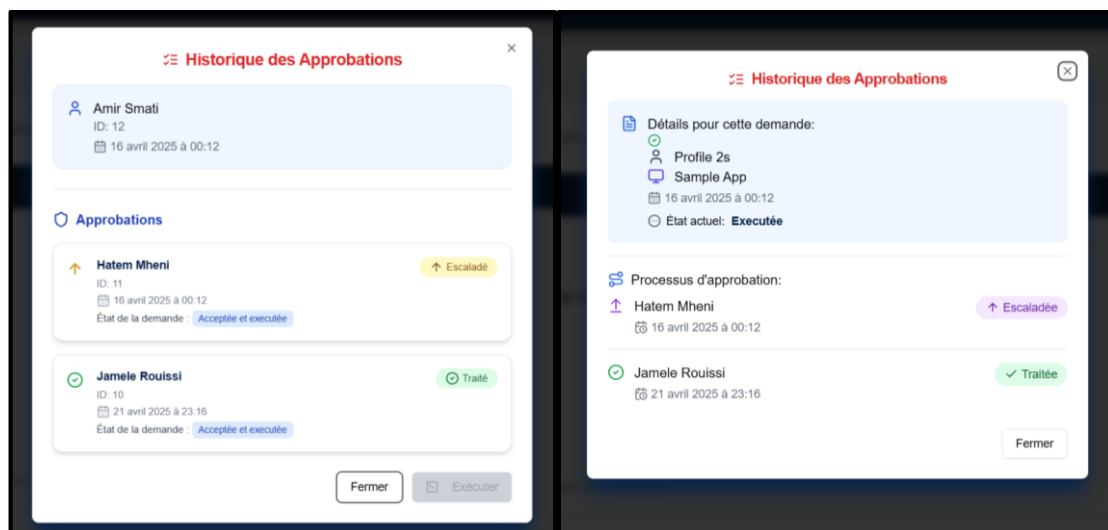


Figure 51: Request Detailed History.

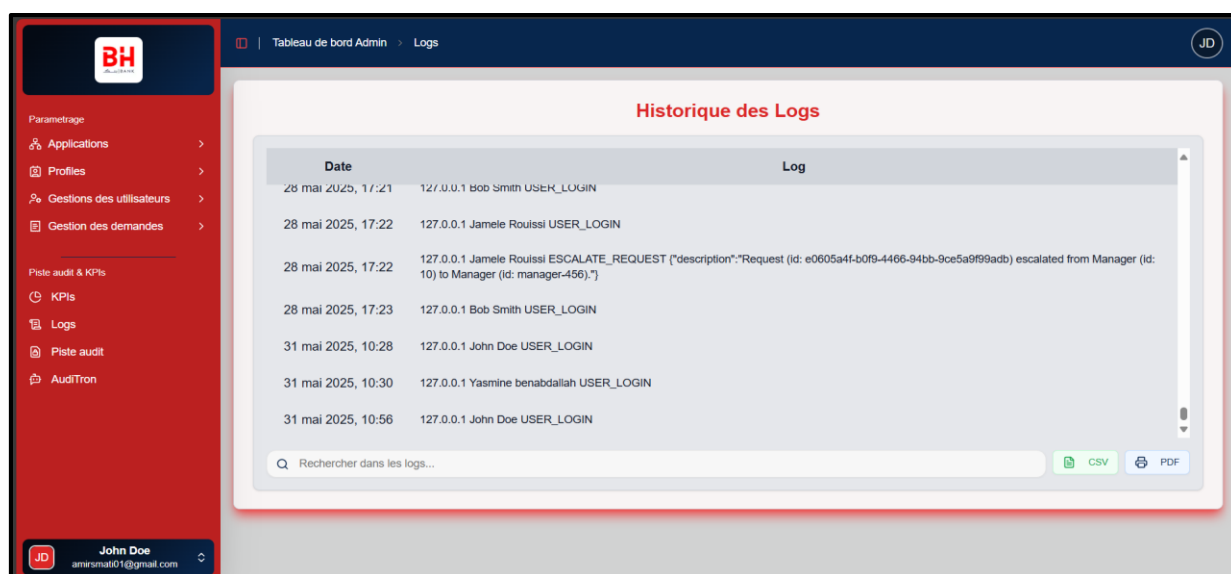


Figure 52: Logs Page.



Figure 53: KPIs Page.

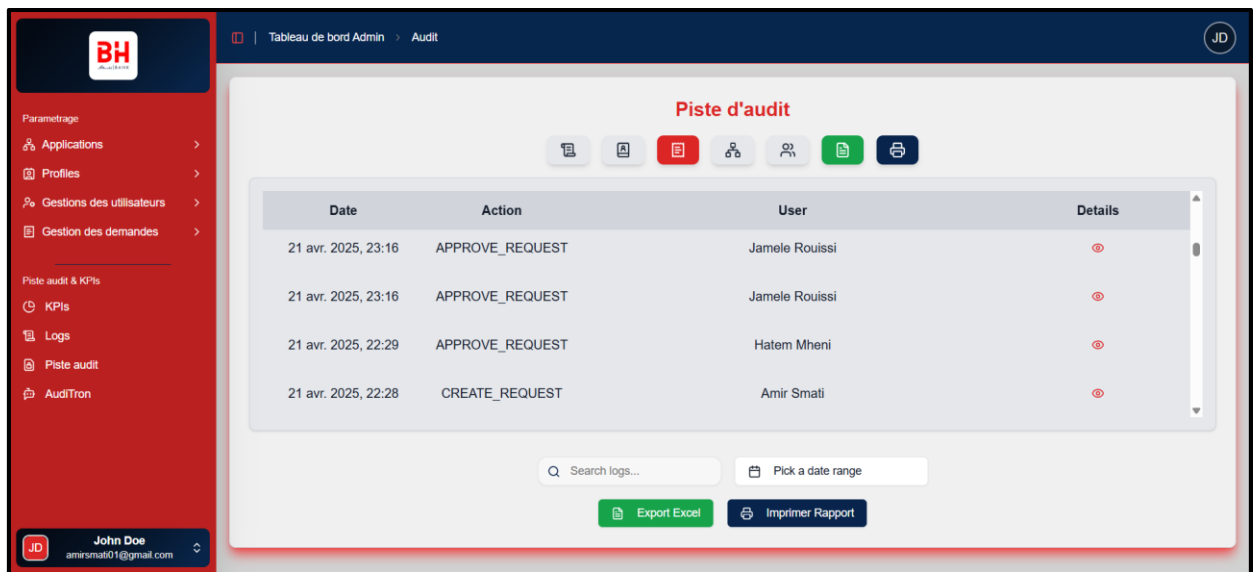


Figure 54: Audit Trail Page.